

# BATTERY HEALTH MONITORING USING MACHINE LEARNING



A Thesis  
Presented to  
The Academic Faculty

By

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## THESIS ACCEPTANCE CERTIFICATE

This is to certify that work in this thesis has been carried out by Mr. **Shehzar Shahzad Sheikh**, (Registration No. 00000204738), student of U.S.-Pakistan Center for Advanced Studies in Energy has been vetted by undersigned and found complete in all respects as per NUST Statutes/ Regulations, is within similarity indices limit and is accepted as partial fulfillment for award of MS/MPhil degree. It is further certified that necessary amendments as pointed out by GEC members of the scholar have also been incorporated in the said thesis.

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When life knocks you down, try to land on your back. Because if you can look up, you  
can get up.

*Les Brown*

I dedicate this thesis to my beloved parents and my siblings, My father Mr. Shahzad Anwar an assistant professor, who taught me trust in Allah and lesson of honesty, My Mother Madam Naheed, who alleviated all hardships and provided me alluring moments of life and my siblings Uzair, Laiba, and Abdul Wasey, who are springs of my life. I also dedicate my work to love of my life Abeera - my fiancée.

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## **Abstract**

Batteries are a combination of electro-chemicals, which empower a large number of devices around us such as electric vehicles and portable electronics. These batteries require proper care using management systems having specialized monitoring features, such as charge controlling mechanisms and temperature regulation to prevent health, safety and property hazards. Therefore, to ensure the regulation of such parameters of a battery, features like state of charge (SOC) and state of health (SOH) become highly necessary. In this thesis, a machine learning-based method is proposed to investigate the aforementioned parameters using a data-driven approach. The estimation of SOH using a simplified machine learning (ML) algorithm that uses feature extraction from the battery discharge curves and obtains the SOH as the outcome of the ML model. With the help of SOH results, The estimation of SOC and SOH relationship of a particular battery have also be established. Extensive simulations have been carried out to study the performance of the proposed method.

### **List of Papers Presented/Accepted**

- **S.S. Sheikh**, M.Kazim, S.Iqbal, and A.Ulasyar, "Real-Time Simulation of Microgrid and Load Behavior Analysis Using FPGA," in *IEEE-iCoMET*, 2019
- **S.S. Sheikh**, "Technical and Economic Challenges of Flow Batteries In Grid-Scale Energy Storages" in *ICRANET-Air University Islamabad*, 2018.
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## Layout of Thesis

This thesis consists of five chapters and each chapter encloses their own summary and bibliography, however, for ease, all references are also included at the end of the thesis.

Here, **first Chapter** Provides introduction of this thesis which includes how this research work is evolved with the help of modern research work in estimation of batteries. Moreover, this chapter also contains **Objectives** of thesis, **limitations** and also provides a thesis map.

**2nd Chapter**, summarizes the literature review and detail knowledge related to batteries specifically to lithium-ion (Li-ion) in terms of electrochemical analysis, comparison between Li-ion and other batteries, and also provides overview of different estimation techniques for battery health estimation. Moreover, this Chapter also provides mathematical modeling of Li-ion and also shares battery discharging curve trends and aging process.

**3rd Chapter**, shares literature on machine learning. Here, machine learning is defined and also shares why machine learning is needed, type of machine learning techniques in terms of their implementations and applications and finally provides insight of the type of data set used for this thesis.

**4th Chapter** describes the methodology used in pursuit of this research work, understanding of Machine learning and data set used in this thesis. which includes flow of algorithms, overview of machine learning technique for state of health (SOH) and calculation of state of charge (SOC) and also advocates what feature must be selected and how features are extracted via NASA data set. moreover this chapter also includes, selection of features, the battery discharge curve's trend and labeling of non-linearity in discharge curves. Moreover, it also provides detail development of mathematical model of knee-point identification algorithm and finally, encloses SOC versus SOH algorithm detail.

**5th Chapter**, contains all results related to SOH of all different batteries such as batteries with different current and temperature levels respectively. This chapter also provides

detail how such results are drawn. and finally, SOC versus SOH result displays an important relationship between charge of battery and degradation process, while providing results, this chapter also provides discussion on the drawn results and shares how such results are important towards contribution in estimation of Li-ion batteries, while keeping a balance between two different domain knowledge, i.e., power engineering and machine learning.

**6th Chapter**, encloses conclusion and future recommendation of this thesis, while suggests that, there always exist room for improvements and such improvements also excites other researchers to carry their research work in battery management system with support of this work rather starting from sketch.

**Appendices** of this thesis contains complex results which provide insight about SOH relationship with features. Moreover, the related work provides how battery internal impedance can be extracted in low cost hardware such as with Arduino.

**References** closes this thesis, these references provides latest research work related to batteries and estimation techniques that are highly helpful for future fellow researchers.

# **CHAPTER 1**

## **INTRODUCTION**

recent decades, global climate has degraded, and greenhouse effect has become prominent due to the rampant increase in earth's temperature by carbon-emitting fossil fueled vehicles and conventional power generation resources. This has globally driven and endorsed the research and development of energy storage systems. In the evolution of several energy storage devices, Lithium-ion (Li-ion) has emerged as one of the best suited and widely implemented solutions with extensive usage in low power portable electronic devices, power sources for electric vehicles and for future grid-scale energy storage [1]. High energy density, high power density, safety, a large number of life-cycles and low self-discharge are the main attributes of Li-ion, which make it superior to other batteries such as lead-acid, Sodium Sulfur (NaS) and Nickel Metal hydride (NiMH) [2].

The electrical characteristics of Li-ion batteries, i.e., energy-density, power-density, and C-rate, have upgraded significantly, albeit the problem remains with the non-linear and non-uniform charging capacity and aging retention with a large number of cells in battery packs [3]. To overcome this inherent pitfall of Li-ion batteries, a battery management system is a prerequisite to protect the overall system and monitor the optimal usage of energy [4]. Such battery management systems need to ensure and monitor the state of functions of batteries including the state of health (SOH) and state of charge (SOC), which in return, optimize the battery performance and life [5],[6], [7]. As an example, in electrical vehicles, the SOH is analogous to an odometer and SOC mimics the fuel gauge used in fueled vehicles. If SOH and SOC are estimated accurately, they can prevent overcharging, overheating and extend battery life, making them highly important parameters [8]. However, they can not be measured directly by any electrical equipment, as a substitute, they are estimated by variables of batteries like input load current (1A and 2A) and terminal voltage at output. In

this regard, many techniques have been developed to estimate SOC and SOH and each has its own advantages and shortfalls [9].

The SOC of a battery shows the presently available charge which changes over the utility of load current drawn from the battery [10]. Present literature classifies the SOC into three broad types of estimation techniques; data-driven approach, adaptive methods, and hybrid techniques [11]. The data-driven approach is further classified into two methods; one of them directly measures the SOC and the other method is model-based [12]. The direct measurement methods include open circuit voltage method (OVC) [13], Coulomb counting method (CCM) [14] and electrochemical impedance spectroscopy (EIS) [15]. The OVC provides accurate SOC but requires relaxation time intervals between two cyclic measurements and is prone to more errors when OVC-SOC has a flat curve. The CCM measures the rate of discharging current and integrates the current and time to find SOC [16]. However, this method depends on the accuracy of the current sensor and due to the open-loop nature of CCM, it accumulates error [17]. The last method of the data-driven approach is model-based and it is an online method which takes waveforms as model computes the output, i.e. the SOC. Two prominent model-based methods have widely been used; the electrical equivalent and the electrochemical models. The biggest shortfall of the model-based method is that they can only be applied on fresh cells for model parameters and are invalid for aged/used cells [18].

According to the best of knowledge, methods like Kalman filter (KF), fuzzy logic (FL), support vector machines (SVM) and machine learning techniques including artificial neural network are categorized in adaptive methods for SOC [19],[20],[21],[22]. Among them, the Kalman filter is the most employed method, which verifies SOC through real-time state estimation with high accuracy. Lastly, hybrid models avail the benefits of both data-driven and adaptive methods such as extended/uncentered Kalman filters.

The SOH defines the health of the battery in terms of aged battery performance versus ideal/new battery. A proxy of SOH is taken as internal resistance, which increases

over the time. However, it is difficult to calculate real-time internal resistance. SOH can also be estimated by two methods; the data driven approach and adaptive methods [23]. In data-driven approach, the data for discharging voltage and load current cycles can be used to estimate SOH [24]. However, this technique requires depth of knowledge and understanding of the correlations between different operations versus aging of battery. The accuracy of this method depends on the size of the dataset; a large authentic data set will yield the best results [25]. Adaptive methods employ only those parameters that are prone to anticipate in aging of the battery. These parameters should be examined carefully as they will be used throughout the operation. There are many techniques to estimate SOH, including Kalman filter, extended Kalman filter, fuzzy logic, intelligent algorithms, e.g., genetic algorithm (GA), artificial neural networks, least square polynomial regression, and particle filters. For prognosis of battery, EKF is commonly used and is easy to implement. Recursive least-squares with electromotive force method is used for online estimation for SOH and SOC. Load data as collectives are used for training and testing in SVM. Battery parameters, diffusion capacitance and the SOH is estimated using GA [26],[27]. The discharging waveform is mapped by using least squared polynomial regression, an ANN is used for mapping features and predicting using a particle filter. However, such techniques provide good efficiency but are computationally expensive in terms of their pre-processing stages. The compounded uncertainties can raise beyond the control in estimation, if not carefully managed [28].

## 1.1 Objectives

In [28], the end of discharge has been estimated by mapping discharge voltage curves over load current using least-squared polynomial regression and artificial neural networks (ANN) to predict the SOH for different loading profiles. However, in their work, many assumptions have been made, i.e., predetermination of the slope values, and lengthened mapping processes for prediction of EOD. whereas these assumptions have been overcome and introduced a method for SOH estimation therefore, the main contribution of this thesis can be summarized as follows:

- *To accurately define knee points (i.e., initial voltage drop and threshold value) and implemented machine learning algorithms with realistic model, whereas, In previous works related to the data driven approaches, many approximations and assumptions related to discharge curves have been made.*
- *To implement the simplified machine learning for SOH. To enable the trained models have the ability to accurately predict and estimate results for different current and temperature levels respectively.*
- *To employed a particular value of voltage, which can not only able to provide the SOC of any discharge cycle correctly but can also have intelligence of drawing a SOC versus SOH relationship.*

## **1.2 Limitation of Thesis**

Despite the milestones objectives have been achieved in this thesis, there always still exists space for improvement, similarly in this thesis the some limitations were concerned with technical difficulties and some related to confidence of this research work. The nature of this research work is interdisciplinary and required experts from different domains such as of communication and machine learning and expert knowledge of power and energy. therefore at first hand, while considering battery estimation using machine learning requires a good amalgamation of supervision. Secondly, there exist an enormous vacuum between power engineering and other technological developments related to Artificial intelligence (AI) based systems in communications and computer sciences which are becoming the necessary skill for advanced smart grid problems, therefore in future this limitation may be reduced with interdisciplinary research of this thesis types.

While technical difficulties in this thesis were related to data limitations. Here 06 batteries have successfully estimated with the help of machine learning, however, the results could have been more improved if there would have been large data sets. dealing with data set of batteries there exit large data sparsity which lower the accuracy and testing results of machine learning. as aforementioned such sparsity could be reduced with large data sets. therefore, this thesis had to limit to simplified machine learning approaches.

### 1.3 Organization of Thesis

The Thesis is organized as follows. The Chapter 2, provides **literature review** on the background on battery characteristics, the battery modeling approach, which includes the mathematical model of the battery. Chapter 3 presents the literature review on machine learning techniques and types, In chapter 4 provides methodology of thesis presents machine learning for SOH and Chapter 5, shares results and discussions. Finally the thesis summarizes the main conclusions and provide future recommendations in chapter 6.

## CHAPTER 2

### LITERATURE REVIEW

#### 2.1 Background on Battery characteristics

Batteries are a converter or transducer of chemical energy into electrical energy, and vice versa. They consist of an anode and a cathode (i.e., electrodes) engrossed in an electrolyte solution. An electrical potential is created in the battery due to the difference in chemical-driven forces between two electrodes in a cell, which is determined by the product of the reaction and the reactants (i.e., standard Gibbs Free energies) [29]. The data used here is of batteries whose OCV voltage reading is observed at  $24^{\circ}C$  and  $4^{\circ}C$ , and at 1 atmospheric pressure or 1 Mole concentration. This OCV voltage is not exactly available at the terminal during the usage of the battery due to the voltage drop by different processes, i.e., resistive forces/components inside the battery (electrolyte), terminal leads, and potential drop[30].

It should be noted that these parameters of the battery are current dependable and can only be observed as the current is drawn from the battery and this curve changes over the amount of time for a given current level as an example, 2A as shown in Fig. 2.1.

Since the current drawn provides such important understanding about the losses inside the batteries, it helps in quantifying the performance of the battery. In literature, the current drawn from the battery is related to the commonly used term as C-rate. The C-rate ( $C/r$ ) is defined as the amount of current drawn in a single discharge cycle of battery for any given number of hours with respect to its nominal capacity [31]. If a battery has a C/10 (0.1 A) for a battery of 1 Ah, then this battery will be able to provide charge for 10 hours. In addition to this, the amount of energy delivered by a battery-related by the area under the discharge curve also relies on the C-Rate. Fig. 2.2 shows different current, i.e., 1A and 2A, and for different temperatures, i.e.,  $24^{\circ}C$  and  $4^{\circ}C$ .

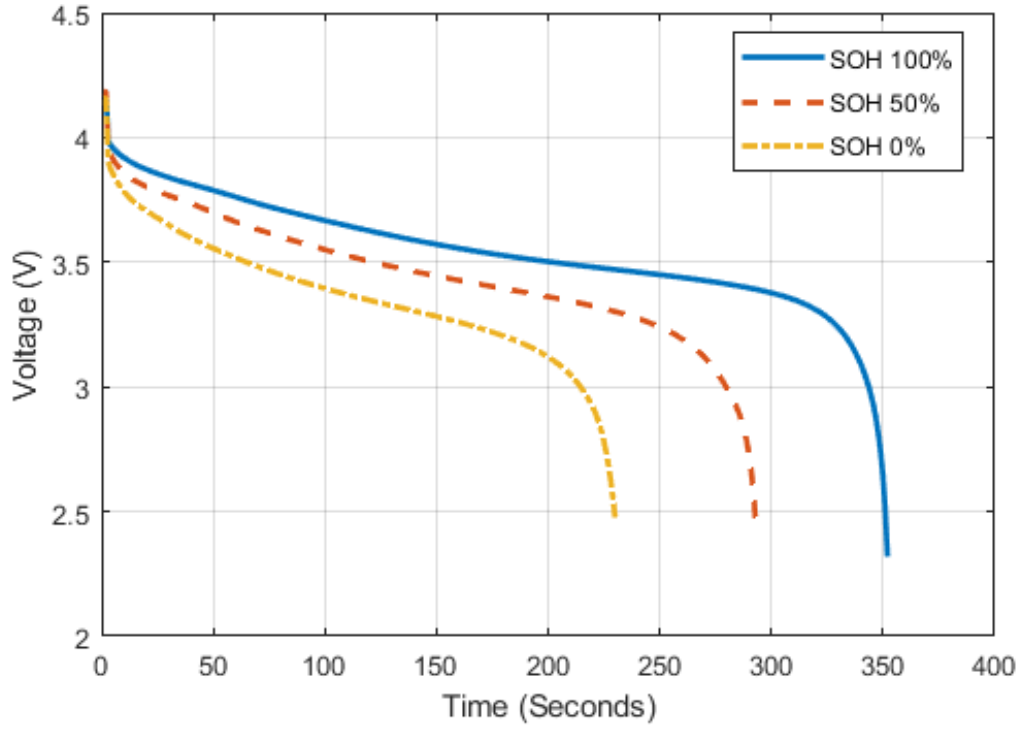


Figure 2.1: Battery discharge curves at constant load current of 2A.

There are some other main important parameters of batteries which represent the physical properties of batteries and based on such parameters, different batteries can be used for different applications such as grid-scale applications to portable electronic devices. Such variables include specific energy, specific power, and cycle life. Specific energy is defined as energy per unit weight or energy stored per unit volume is called energy density. This is an important parameter for the application of portable electronics and devices where the long-lasting charge is required. Specific power (power per weight) or power per volume is defined as power density and this parameter of a battery is important when sudden charge/discharge is required such as in EVs brakes[32]. Lastly, battery cycle life defines how many charging/discharging cycles a battery can provide before it depletes beyond acceptable (normal) limits, i.e., almost 20%. Table 2.1 shows variables of different batteries currently available in market.

Table 2.1: Market-Competitive Batteries and Their Variables.

| <b>Type</b>            | <b>Lead-Acid</b> | <b>NiMH*</b> | <b>NaS*</b> | <b>VRB*</b> | <b>Li-ion</b> |
|------------------------|------------------|--------------|-------------|-------------|---------------|
| Cycle Life             | 200-1000         | 180-2000     | 2500-4000   | 12000       | 1000-10,000   |
| Energy Density (Wh/Kg) | 25-50            | 60-120       | 150-240     | 10-30       | 75 - 200      |
| Power Density (W/Kg)   | 75-300           | 250-1000     | 150-230     | 80-150      | 500-2500      |
| Capital Cost (USD/KWh) | 100-300          | 900-3500     | 300-500     | 150-1000    | 300-2500      |
| Round-Trip Efficiency  | 75-85            | ~ 65         | 75-90       | 75-90       | 85-97         |
| Self-Discharge         | Low              | Medium       | -           | Negligible  | High          |

\*NiMH: Nickel Metal Hydride; \*NaS: Sodium Sulfur and \*VRB: Vanadium Redox Battery

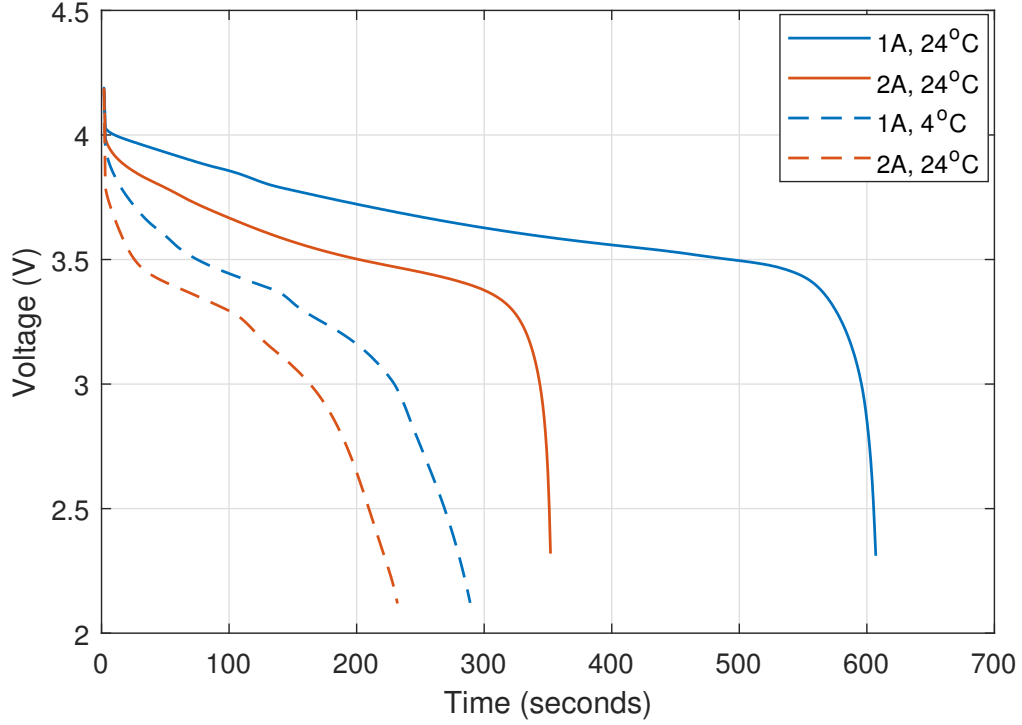


Figure 2.2: Discharge curves for different current, (i.e., 1A and 2A) and for different temperatures ( $24^{\circ}\text{C}$  and  $4^{\circ}\text{C}$ ).

## 2.2 Li-ion battery properties

The Li-ion batteries are the advanced rechargeable batteries first commercially manufactured and developed by Sony in the 1990s. besides Li-ion there are several other rechargeable battery technologies including Lead-acid Nickel-cadmium (NiCd), Sodium Sulfur (NaS), and Redox Flow batteries and etc., which each having prominent characteristics, however, Li-ion is widely used celebrity for its variety of applications including consumer electronics devices to high power equipment, to EVs, even for grid-scale to space applications. Some of the main merits of Li-ion, which make it superior to other competing technologies are listed here which can also be shown in table 1.

A Randles model which is consisting of an electrical Resistor  $R_{\infty}$ , series of parallel resistors and capacitors, and  $C_w$  as Capacitor, however in our model we have used simplified form of 1<sup>st</sup> order  $RC$  model for our analysis that contains  $R_{\infty}$ , a parallel resistor

and Capacitor representing charge carriers conduction of electrolyte, and double layer capacitance, and charge transfer resistance, respectively as shown in Figure.1. This model is implemented in *SimuLink* environment of *Matlab*.

Compare to other types of batteries e.g., NiCd and lead-acid, Li-ion batteries are lightweight and can store more in a given weight per Kg. lithium itself is a highly reactive substance which makes it stores more energy in its atomic bonds hence Li-ion has high energy density and high efficiency as compared to other technologies.

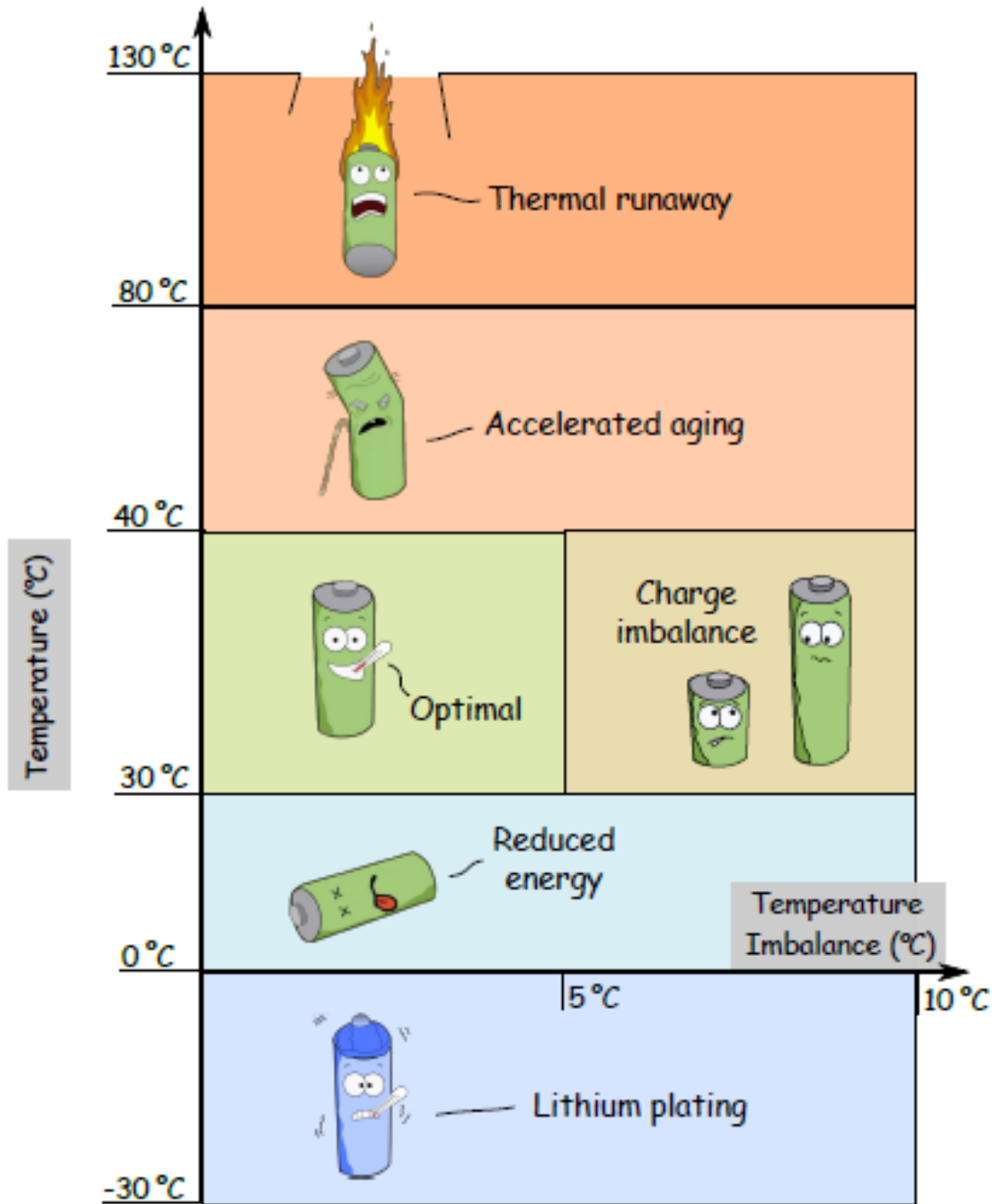


Figure 2.3: Battery behavior at different temperature levels. [32]

- *Self-discharge is defined as the discharge due to the flow of residual ions and electronics in a cell even when there is no current externally drawn from the battery. Higher the self-discharge lowers the period for which battery can store charge. Li-ion has low self-discharge and high charge retention capability as compared to other*

batteries, e.g., NIMH.

- *Li-ion battery has no memory effect as compare to other batteries like NiCd. Memory effect means a battery has been completely discharged before new recharging to hold its full charge capacity.*
- *Li-ion has long cycle life which means they can be operated for a large number of charging and discharging cycles before they should be discarded ( 2,500 -4,000 cycles). However, Li-ion has some of their notable disadvantages as well.*
- *Li-ion batteries are most prone to high temperatures than most other battery chemistry. They degrade rampantly (faster than the normal) if they exposed to extreme temperature (hot) and operating conditions. This property of temperature sensitivity is also known as thermal runaway.*
- *Li-ion life degrades even if they are not used for any application or shelf-life decrease with time, which means as they are manufactured they are degrading whether or not they are used. This degradation increases internal resistance that causes capacity fading, unlike other chemistry.*
- *Li-ion can heavily be damaged if they go under deep discharging, i.e., discharging below the manufacturer recommended threshold voltage (in our case below 2.7 V for a cell), due to this, on-board circuits are installed in Li-ion battery packs which make them costlier than they already are.*
- *During an application of Li-ion anode can cause heating and cathode produce oxygen while lithium itself is highly reactive which possibly lead the battery to explode or get burned/fire. This makes they li-ion battery less safe as compared to other batteries, i.e., NiCd*

Such merits and limitations help us understand their optimal utility for a different application. Therefore, to better understand the complex internal structure of Li-ion, modeling

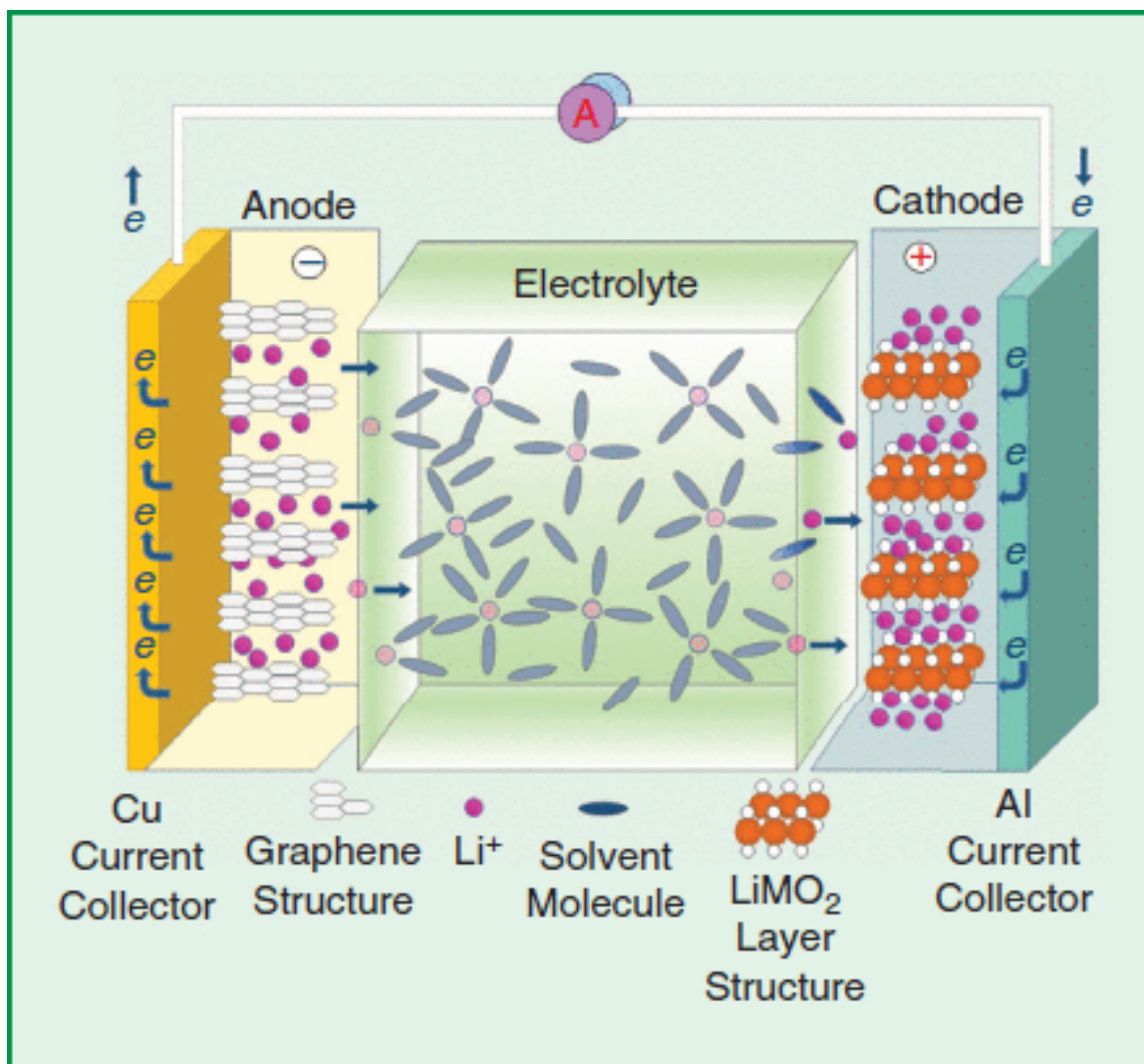


Figure 2.4: Electrochemical Representation of the Li-ion Cell Structure. [2]

and mathematical based approach has been used to explain li-ion chemistry .

### 2.3 Battery Modeling Approach

There are several modeling approaches which have been used to model internal characteristics of Li-ion which can be broadly categorized into three types, i.e., black, gray and white box models[2]. The first principal method is categorized as a white-box model which is developed with internal battery dynamics such as intercalation kinetics, ion diffusions, and electric potentials, however, the internal electrochemical reactions can be very dull and

computationally intractable due to initially partial-differential equations usage [33]. Another approach is to use model-based algorithms which are merged with low-order modeling and relies on the slight sacrifice of precision and physical representation[34]. Furthermore, an alternative method is data-driven or can also be called model-free approaches where machine learning techniques can be used. A gray/black box model referred to as electrical equivalent circuit model [35] in which internal characteristics of a battery is modeled in terms of resistors, capacitors, and inductors. Here we have used the lumped model with approximation of  $IR$  drop (i.e., caused by electrolyte resistance) is represented as  $R_o$ , the activation polarization is denoted as a charge transfer resistance  $R_p$  and  $C_p$  is modeled as dual layer capacitance in parallel with  $R_p$  as shown in Fig. 2.5.

### 2.3.1 Mathematical Model of a Battery

The battery characteristics can be identified using Randles models which is one of suitable model for identifiability of battery dynamics, i.e., internal impedance [36]. The generalized Randles models have also been used in bio-technologies including bio-medical where a pair of  $RC$  and voltage potential is modeled for electrode-tissues connecting with electroencephalography [37]. In [36] has identified the identifiability of Randle models where a Randles model consists of an electrical Resistor  $R_o$  representing an ohmic resistance (i.e.,  $IR$  Drop) due to conduction of charge carriers through electrolyte and metallic conductions, series of parallel resistors and capacitors (i.e., activation polarization) represent charge transfer resistance and double layer capacitance, represents diffusion process, respectively [38],[39]. The number of  $RC$  parallel circuits defines the order that reduces the error between developed model and real-time measure voltages [40], A second order Thevenin equivalent circuit model (i.e., Dual Polarization (DP)) since this DP model stands satisfactory for our case as shown in Fig. 2.5.

The electrical behavior of the circuit is given by following equations:

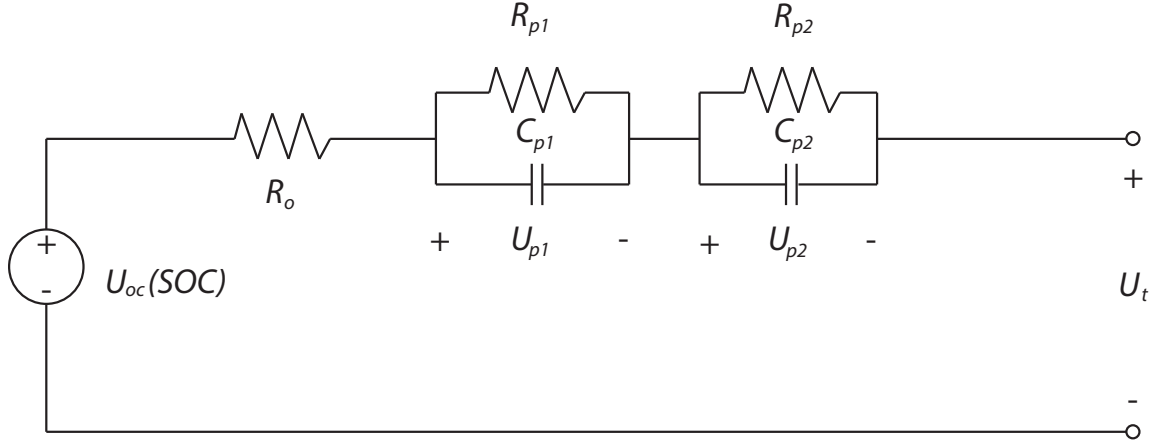


Figure 2.5: Electric equivalent model of Lithium-ion cell.

$$\frac{dU_{p1}}{dt} = \frac{I_t}{C_{p1}} - \frac{U_{p1}}{R_{p1}C_{p1}}. \quad (2.1)$$

$$\frac{dU_{p2}}{dt} = \frac{I_t}{C_{p2}} - \frac{U_{p2}}{R_{p2}C_{p2}}. \quad (2.2)$$

$$\frac{dSOC}{dt} = \eta \times \frac{I_t}{C_{batt}}. \quad (2.3)$$

The output equation of the DP model according to Fig. 2.5 is

$$y = U_{oc}(SOH) - U_{p1} - U_{p2} - R_o \times I_t. \quad (2.4)$$

Whereas,  $U_{oc}(SOC)$  is internal voltage of battery,  $U_{p1}$  and  $U_{p2}$  are voltage across  $RC$  parallel branches.

### 2.3.2 Randles Models Mathematics

The general mathematical parameterization of Randle model is explained in [36], which has been used for 1<sup>st</sup> order  $RC$  model in this case. In implementation of such model, State-space and Transfer function for generalized Randles circuit are derived.

Here system input is fed as electric current  $u(t) = i(t) \in R$ , whereas voltage is taken as  $v(t) \in R$ . the  $RC$ 's parallel branches voltage as  $x(t) = [v_1(t), v_1(t) \cdots v_n(t), v_w(t)]^T \in R^{n+1}$ , and  $\theta$  parameter set of model containing open subset of  $\mathcal{D} \in R^{2n+2}$ .

$$\theta = [R_\infty \ R_1 \cdots R_n \ C_1]. \quad (2.5)$$

## **2.4 Summary**

This chapter summarized the literature review and detail knowledge related to batteries specifically to lithium-ion (Li-ion) in terms of electrochemical analysis, comparison between Li-ion and other batteries, and also provided overview of different estimation techniques for battery health estimation. Moreover, this Chapter also provided mathematical modeling of Li-ion and also shared battery discharging curve trends and aging process.

## **CHAPTER 3**

### **LITERATURE REVIEW ON MACHINE LEARNING**

#### **3.1 Definition of Machine learning**

Machine learning is a concept to learn from experience and examples, without getting improved explicitly in programming or writing codes, one only need to feed data to a machine learning algorithm and it will build a logic based on the data available. Moreover, Machine learning can be classified into two board discussions as shown in Fig. 3.1, one involves the functionality of machine learning with respect to its application such as, algorithm for classification that help in clustering desire data sets. Moreover, it can also be understood by examples of classification of handwritten alphabets, which could be classify as which exact letter is related to class of alphabets, or identification of voice, classification of spam and not-spam messages, facial recognition, email filtering, medical diagnosis and selection between good and bad products in any production line are common examples of classification. Moreover, functionality based on machine learning algorithm can be used for prediction purposes such as, prediction of weather, prediction of human health over the passage of time and age groups.

While categorizing the date set into their desire fields, domain and classes, machine learning helps in prediction of data trends as example stock exchange in finance, prediction of human health in upcoming months or even development of cancer cells in human body are examples of prediction, moreover, prediction can easily to comprehend by the examples of how much profit can be obtained with a given yield.

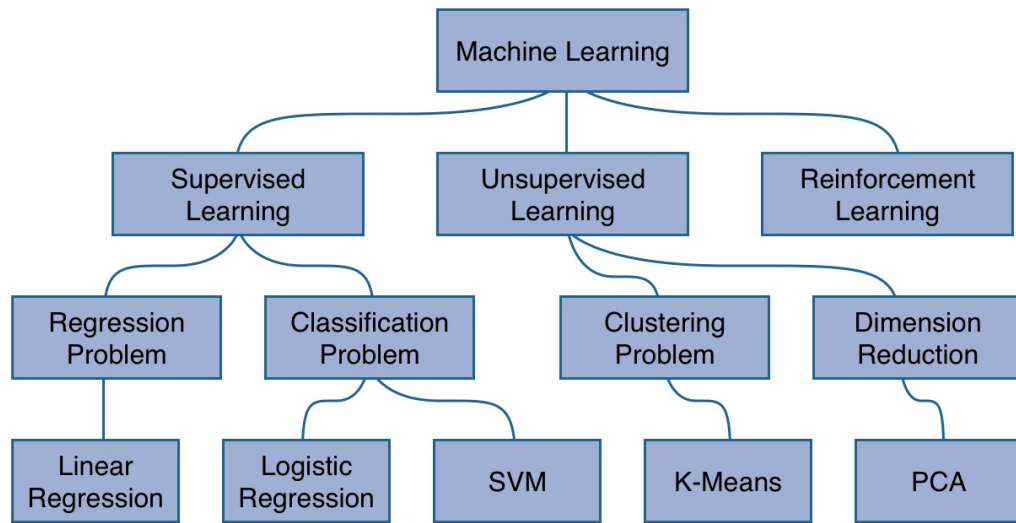


Figure 3.1: Commonly used Machine learning Hierarchy.

### 3.2 Need of Machine Learning

Machine learning is sub domain of Artificial intelligence (AI) which is emerged out of it. With AI the goal always remains to improve to develop more and more intelligence in machine such as computers or other equipment. While considering AI, the expectation remains high with complex tasks as example delivering an amazon package with the help of drone in bad weather. This may seem easy to make it done, however there involves complex algorithms which not only deliver the package on the exact home but also manage to fly in stormy winds and rains which highly affects the path ideally chosen path. There exist some intelligence which not only deliver package to ones door step. While considering this scenario there operates many advanced algorithms such as adoptive path identification, torque adjustment of drone motors and object detection, etc. each such task requires series of machine learning algorithms, i.e., for adoptive path or track identification there may requires prediction and estimation of current and future states, similarly for torque adjustment there may requires weather forecast data and during the flight of drone online path correction with the help of linear logistic regression algorithm or any advanced algorithm of machine

learning and for object detection such as ground or any other obstacle there may also exist imaging processing algorithms.

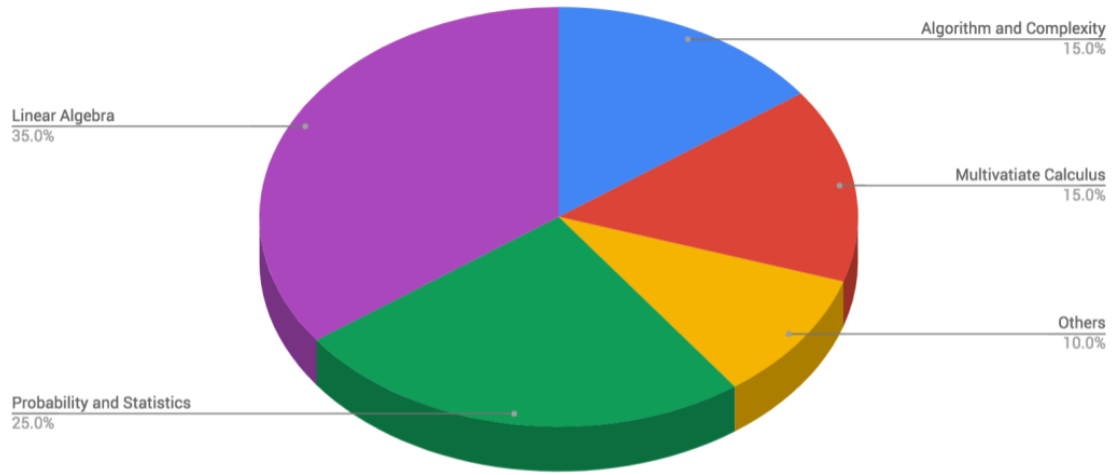


Figure 3.2: Importance of Mathematics topics needed for machine learning.

Finding patterns in data is only possible for human mental capacity, however with the increment of data human brains require more processing time to perceive it, whereas machine learning can easily understand such patterns with minimum time. Similarly, with increased usage of Internet of things (IoT) in daily life, each device is now connected with internet facility therefore at the same place where IoT provides ease to human beings there arise challenge of such data handling for data-scientists and organization to keep a track on their data. Similarly, with in the modern grid, i.e., smart grid facilities integration of large renewable sources from unlimited numbers of nodes from industrial, independent power provider (IPP) and even from domestic users. At the same point, where such large infrastructure facilities and introduces green energy concept and reduces carbon footprints, on the other hand its highly complex to manage such a large energy edifice which may introduce unpredicted contingencies and even blackouts at their extremes. Therefore, there requires machine learning which not only ease data handling by clustering and estimating them into desire domains and classes but also reduce unnecessary data. Machine learning is obedient of data sets and the performance of machine learning algorithm dramatically improves with large data sets, therefore, machine learning has provided groundbreaking

results in the field of medical especially in cancer studies.

”Two definitions of Machine Learning are offered. Arthur Samuel described it as: ”the field of study that gives computers the ability to learn without being explicitly programmed.”

”A computer program is said to learn from experience  $E$  with respect to some class of tasks  $T$  and performance measure  $P$ , if its performance at tasks in  $T$ , as measured by  $P$ , improves with experience  $E$ .” - Tom Mitchell Whereas playing checkers is taken as example,  $E$  and  $T$  denotes the experience of playing many games of checkers and pre the task of playing checkers respectively and finally,  $P$  represents the probability that the program will win the next game.”

### **3.3 Type of Machine learning**

The machine learning is generally categorized into three broad type on the basics of algorithms and applications.

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

#### 3.3.1 Supervised Learning

Supervised learning has major contribution in machine learning and one of the most used learning method. In supervised learning, the machine/system efforts to learn (estimate, predict) from previous experiences and examples. In other words, in supervised machine learning system is already provided with input ( $x$ ) and output ( $y$ ) and an algorithm is develop to correlate relationship between  $x$  and  $y$  as mapping. Similarly in this research thesis has been practiced with data set of NASA, here the data set is randomly divided into two groups; one group is used for training of machine and other group is used to test and

validate the developed algorithm performance, it should also be noted that, in this thesis the 70% of randomly taken data is trained and remaining 30% is tested.

$$h_{\theta}(x_1, x_2) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2. \quad (3.1)$$

While implementing supervised learning, it is important to note that this type of machine learning performance well for medium scale of data sets, if the data set increases beyond a limit, there start to arise issues into it and the main issue would be of data handling. Supervised learning consists of many algorithms of statics as example of linear regression model and artificial neural networks (ANN). While implementing linear regression there are some affiliated concepts of logistic regression and cost functions which defines how computationally expensive a algorithm and how much time a algorithm will take to coverage to its desire value, i.e., optima, furthermore in the details of cost function, concept of global and local optima; a global optima is the overall representation of a function, whereas local optima can be 2 or upto  $(2 + n)$ , Where  $n$  is any positive integer.

$$J(\theta) = J(\theta_0, \dots, \theta_6) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x_1^{(i)}, x_2^{(i)}) - y_m)^2. \quad (3.2)$$

where  $J$  is the cost or error function,  $m$  is the size of the data set and  $\theta$  is the vector containing model parameters  $\theta_0, \dots, \theta_6$ .

Finally, as summary, supervised learning problem can be broadly categorized into two types, Regression and classification.

### 3.3.2 Unsupervised Learning

In unsupervised learning, the system takes its own data-driven decision with only input data set. As name is obvious, the algorithm are only provided with input data ( $x$ ) with no output variables whereas, in supervised learning input and output variables are feed to system. On the bases of data set algorithm decide weather what possible outcomes can be drawn. a

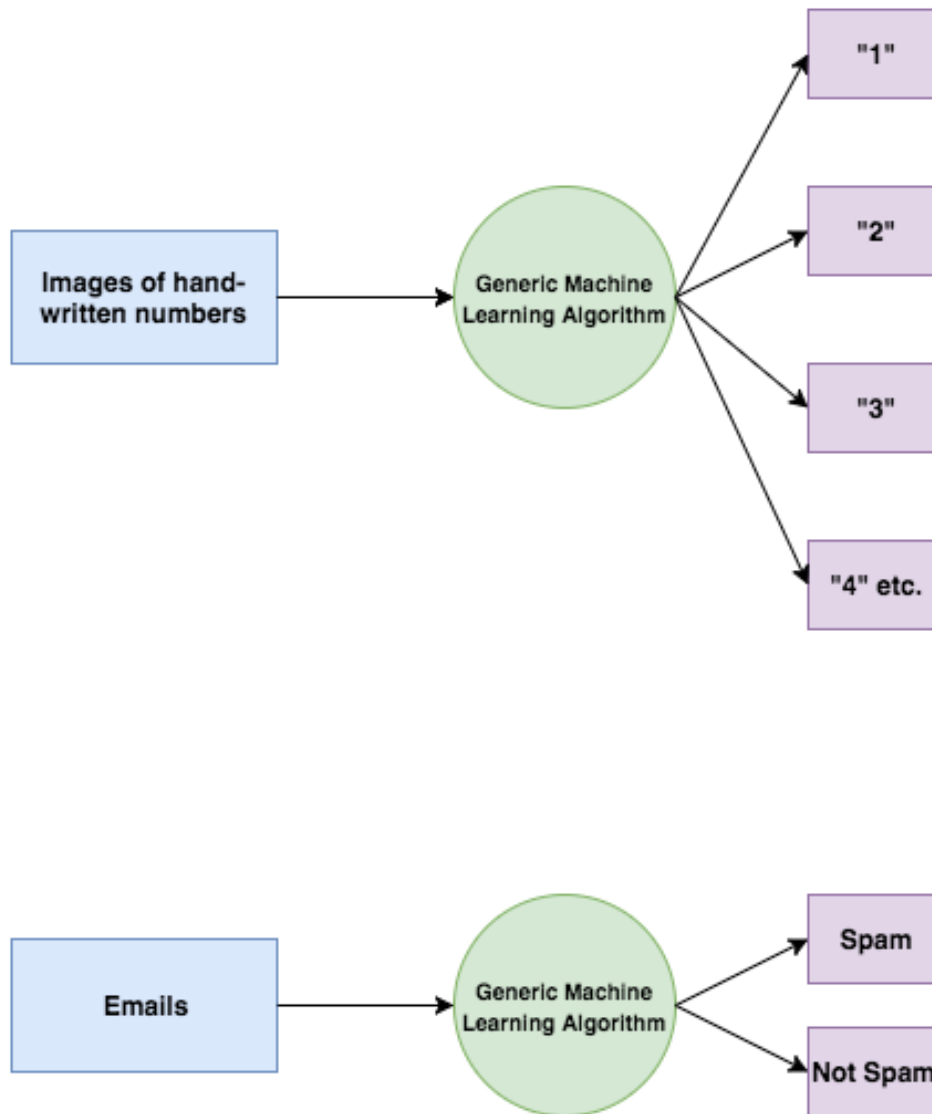


Figure 3.3: Machine learning algorithm for classification of Email as an example.

simplified example is shown in Fig. 3.4 for the classification of unlabeled fruits data and machine classify them recording to its learning.

Moreover, unsupervised learning can be classified into two main categories; association and clustering. Let this be understood by examples. In the case of association, YouTube

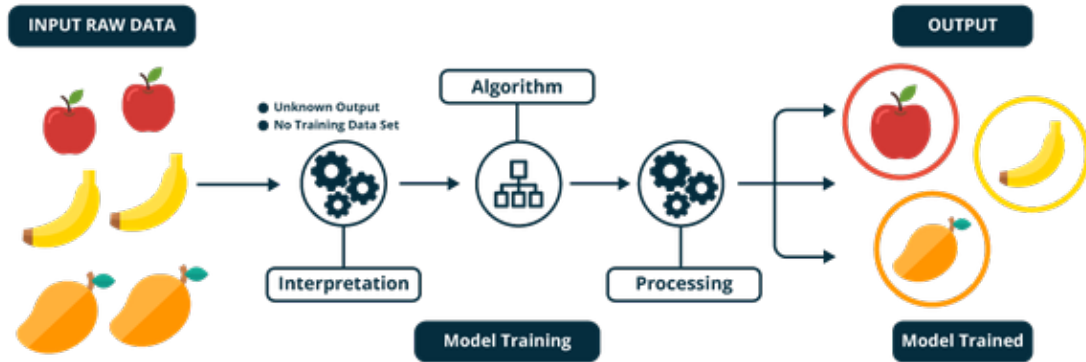


Figure 3.4: A simplified intuition for unsupervised machine learning [44].

can be considered, on the bases of user experiences towards videos, YouTube suggest more options related to user likeness. Therefore, if a user watches ( $Q$ ) videos tend to have interest in watching ( $U$ ) videos. Similarly, with news-websites or application where application associate ( $n$ ) news on the bases of ( $m$ ) news. While examples of clustering includes news clustering. Where in background there are no given output by editors of news or authors, the Google search engine classify news on the bases of data sets, e.g., such search engine clusters keywords, i.e., war, love, weather, child, education and etc. Unsupervised learning algorithms are convolutional neural networks (CNN) or transfer learning which have ability to classify large amount of data. with the passage of time, the usage of unsupervised learning is increasing and as this type of algorithms contains more intelligence.

### 3.3.3 Reinforcement Learning

In reinforcement learning algorithm interact with change in surrounding in with a algorithm is assigned a goal or task. The feedback of such algorithms given feedback in terms of rewards and punishments. Moreover, such algorithms makes decision on the base of trained data as shown by Fig. 3.5.

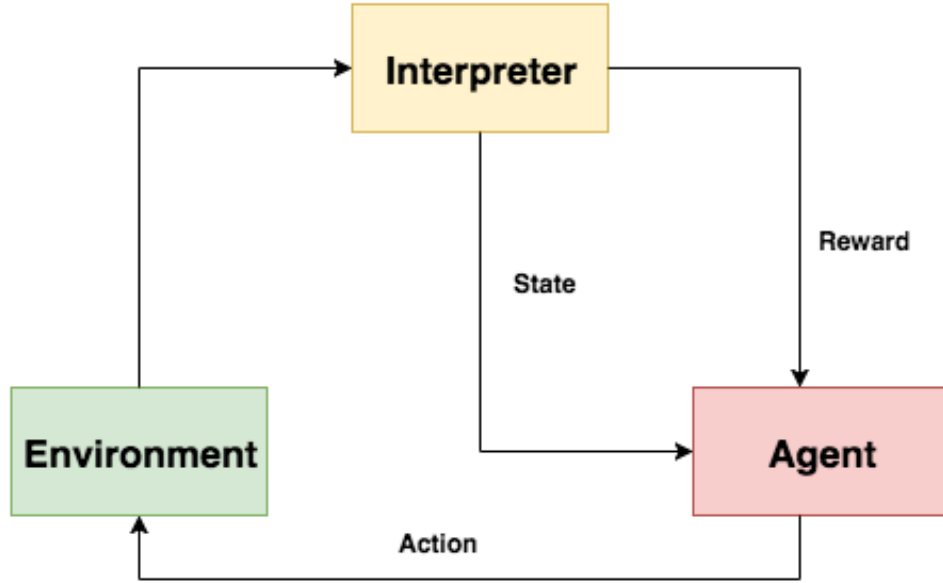


Figure 3.5: Overall layout of Reinforcement learning.

### 3.4 Data Set

In this thesis work, NASA Ames prognostic dataset [41] is used, which is open source data set. While working on NASA data set, it is important to note that there exit different type of battery data set depending upon their current and temperature differences. moreover, on their data respository there are two main type of data set related to batteries, first type is of constant current load profile, as example, a battery is discharge for constant 1A, 2A and 4A, while keeping temperature level at  $24^{\circ}C$ ,  $4^{\circ}C$  and  $44^{\circ}C$ . The detail information is given as below:

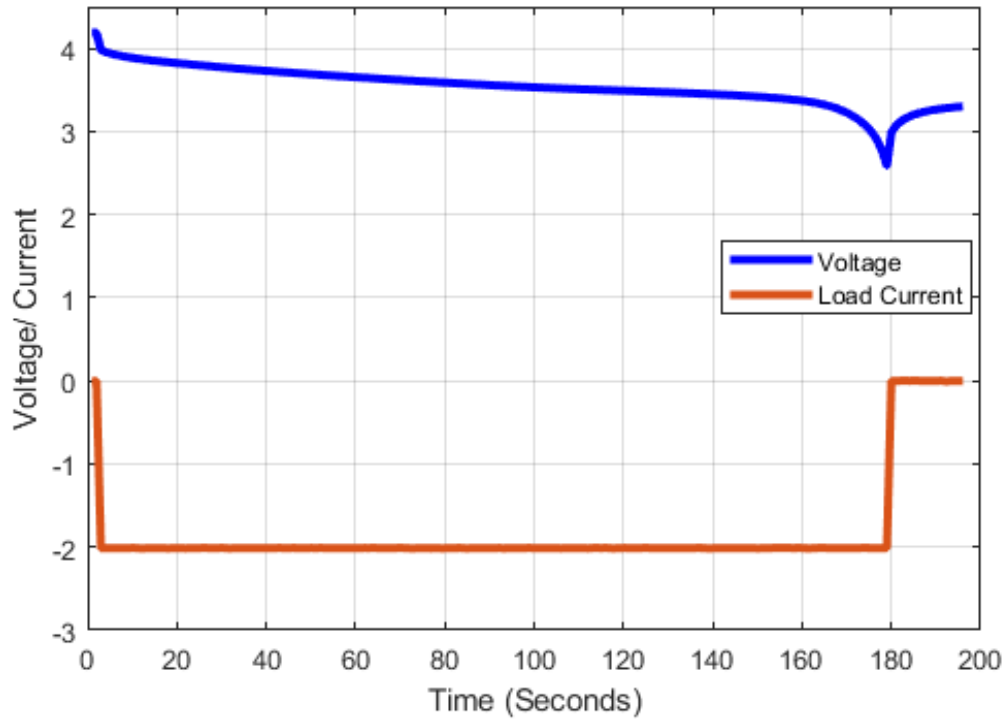


Figure 3.6: A discharge cycle with the voltage and load current.

#### 3.4.1 Batteries at Room Temperature

The data set of first category consists of three batteries such as B0005, B0006 and B0007. These batteries profiles are drawn at room temperature while profiles means charging, discharging and impedance. batteries were charged at constant load profile such as constant current (CC) mode at 1.5A until the battery voltage reached 4.2V and then continued in a constant voltage (CV) mode until the charge current dropped to 20mA. Whereas, the Discharging of these batteries was carried out at a constant current (CC) level of 2A until the battery voltage fell to 2.7V, 2.5V, 2.2V and 2.5V for batteries 5 6 7 and 18 respectively.

Impedance values were found with the help of electrochemical impedance spectroscopy (EIS) with frequency of 0.1Hz to 5kHz sweep. Impedance measurement was carried out through an electrochemical impedance spectroscopy (EIS) frequency sweep from 0.1Hz to 5kHz. repetition of charging and discharging cycles causes aging processing in batteries which can be observed in change of impedance value over the time, such change in

impedance suggest a proxy standard for aging processes. The experiments were stopped when the batteries reached end-of-life (EOL) criteria, which was a 30% fade in rated capacity (from 2Ahr to 1.4Ahr). This data set can be used for the prediction of both remaining charge (for a given discharge cycle) and remaining useful life (RUL).

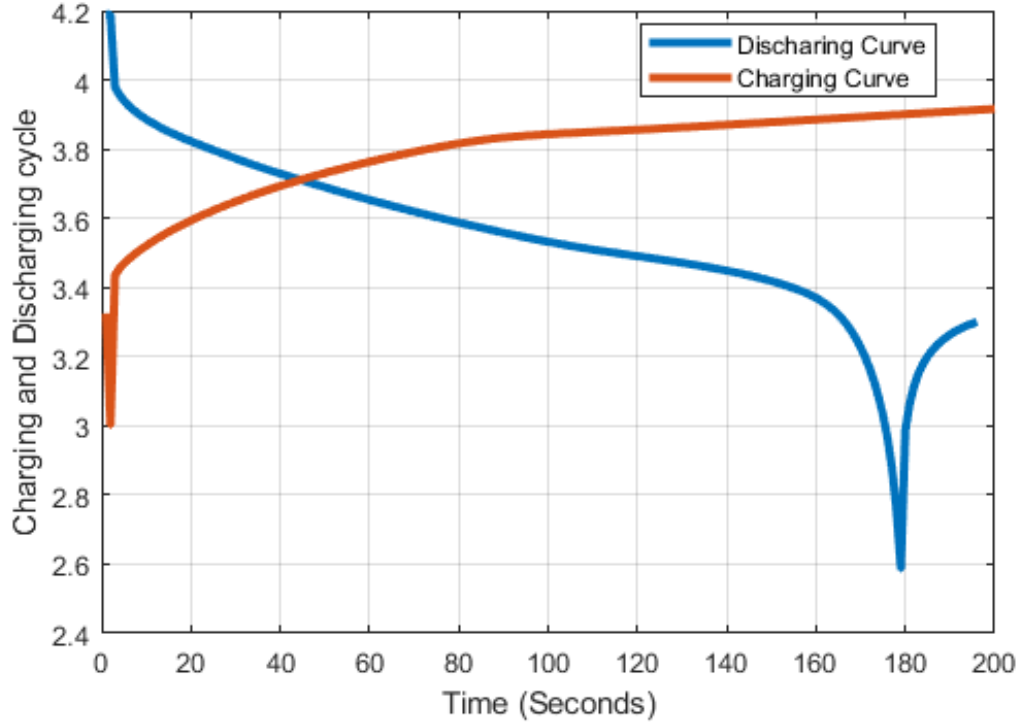


Figure 3.7: A complete cycle of charging and discharging of a battery.

### 3.4.2 Batteries with low temperature

While considering second type of Lithium-ion battery data set which consist of three batteries namely, B0045, B0046 and B0047. similarly, their profiles are drawn with the help of charging, discharging and impedance value. Charging was carried out in a constant current (CC) mode at 1.5A until the battery voltage reached 4.2V and then continued in a constant voltage (CV) mode until the charge current dropped to 20mA. Fixed load current level of 1A was used and the discharge runs were stopped at 2V, 2.2V, 2.5V and 2.7V for batteries 45, 46, 47 and 48 respectively. Impedance values were found with the help of electrochem-

ical impedance spectroscopy (EIS) with frequency of 0.1Hz to 5kHz sweep. Impedance measurement was carried out through an electrochemical impedance spectroscopy (EIS) frequency sweep from 0.1Hz to 5kHz. repetition of charging and discharging cycles causes aging processing in batteries which can be observed in change of impedance value over the time, such change in impedance suggest a proxy standard for aging processes. The experiments were stopped when the batteries reached end-of-life (EOL) criteria, which was a 30% fade in rated capacity (from 2Ahr to 1.4Ahr). This dataset can be used for the prediction of both remaining charge (for a given discharge cycle) and remaining useful life (RUL).

### **3.5 Software platform**

As the NASA data set is available in the form of *structure* type in Matlab therefore it was more convenient to continue working on Matlab. therefore for the development purposes the Matlab is used where extensive simulations were carried out for estimation and data labeling. Moreover, Matlab environment provides great compatibility and commonly used software platform for power engineers and researchers. while working on Matlab the data set were first de-structured to fetch useful information such as discharging curve cycles and other necessary details.

Furthermore, the Matlab provides comfortable interface for machine learning with the help of build-in libraries and applications. in thesis research work, polynomial regression model is implemented with the help of curve fitting tool library.

### 3.6 Summary

**This Chapter**, described the methodology used in pursuit of this research work, understanding of Machine learning and data set used in this thesis. which includes flow of algorithms, overview of machine learning technique for state of health (SOH) and calculation of state of charge (SOC). this chapter also advocated what feature must be selected and how features are extracted via NASA data set. Furthermore, this chapter included, selection of features, the battery discharge curve's trend and labeling of non-linearity in discharge curves. Moreover, it also provided detail development of mathematical model of knee-point identification algorithm and finally, enclosed SOC versus SOH algorithm detail.

## CHAPTER 4

### ESTIMATION OF SOH AND SOC

The machine learning algorithms is implemented for a robust estimation of the SOH and SOC of a battery. Our goal is to calculate the SOH and SOC of a particular battery using a trained regression model, given the battery's discharge curve and the current voltage-of-operation. The SOH is determined using a machine learning algorithm trained on the discharge curves provided by the NASA Ames prognostic dataset [41] which is shown in Table A.1 for different current levels, i.e., 1A,2A and for different temperature levels, i.e.,  $24^{\circ}C$  and  $4^{\circ}C$ .

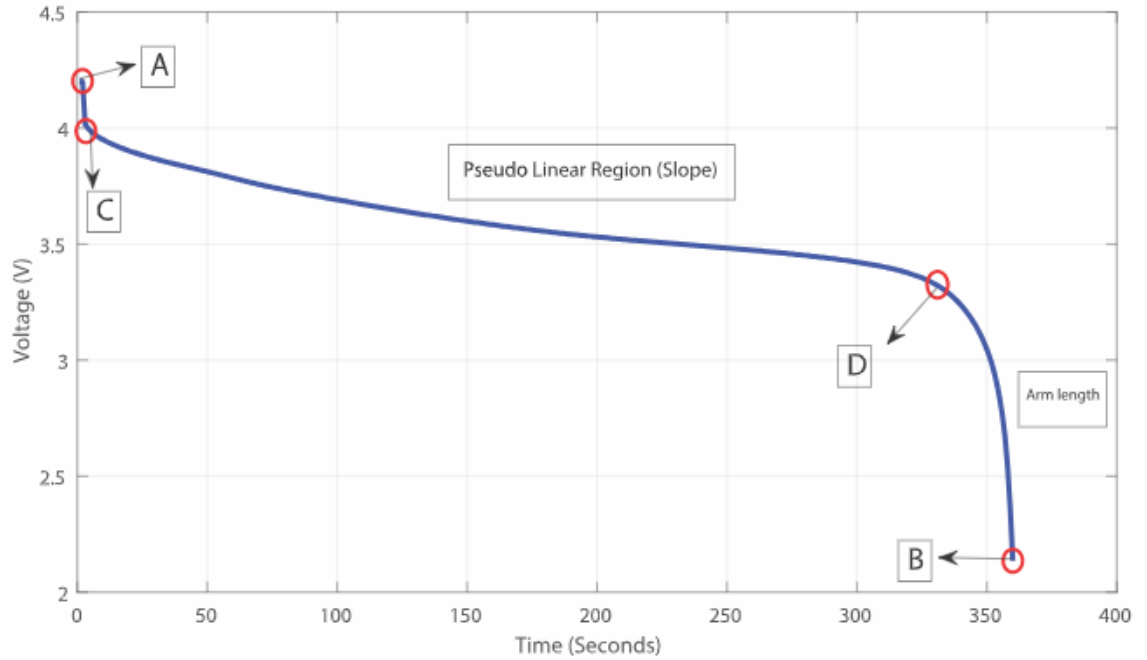


Figure 4.1: Identification of non-linear points.

This model will "look" at a discharge curve and determine its SOH percentage relative to a "healthy" battery. This healthy state of the battery is assumed to be the state at the first discharge in the dataset. The SOC, on the other hand, at a particular voltage-of-operation

Table 4.1: Batteries with Their Current and Temperature Levels.

| S/no. | Battery | Current | Temperature   |
|-------|---------|---------|---------------|
| 1     | B0005   | 2A      | $24^{\circ}C$ |
| 2     | B0006   | 2A      | $24^{\circ}C$ |
| 3     | B0007   | 2A      | $24^{\circ}C$ |
| 4     | B0045   | 1A      | $4^{\circ}C$  |
| 5     | B0046   | 1A      | $4^{\circ}C$  |
| 6     | B0047   | 1A      | $4^{\circ}C$  |

( $V_o$ ) is determined by utilizing the nature of the discharge curve. The batteries have been discharged from a maximum ( $V_{max}$ ) to a minimum ( $V_{min}$ ) voltage. We estimate the SOC by calculating the area under the discharge curve from  $V_o$  to  $V_{min}$  and dividing this area by the total area under the curve. This provides a quantitative percentage measure of how much charge is left in the battery at the particular voltage-of-operation.

This SOC is then plotted against SOH to get a one-to-one mapping. The end-result is the estimation of SOC given SOH at a specific voltage-of-operation. The following subsection details on the evaluation of SOH and SOC.

This SOC is then plotted against SOH to get a one-to-one mapping. The end-result is the estimation of SOC given SOH at a specific voltage-of-operation. The following subsection details on the evaluation of SOH and SOC.

#### 4.1 Knee-point Calculation

The first task is to convert the discharge voltage vectors into measurable and logical features for SOH estimation. An exemplary discharge curve is shown in Fig. 4.1, where points A and B show the maximum voltage ( $V_{max}$ ) and minimum voltage ( $V_{min}$ ) of the discharge curve, respectively. On the other hand, points C and D show the two knee-points of the curve. The relative health of the battery decreases with a decrease in the distance between the two knee-points. Hence, our first input parameter becomes the pseudo linear region between the two knee-points of the discharge curve.

We utilize an extended version of the algorithm proposed in the IBM data platform

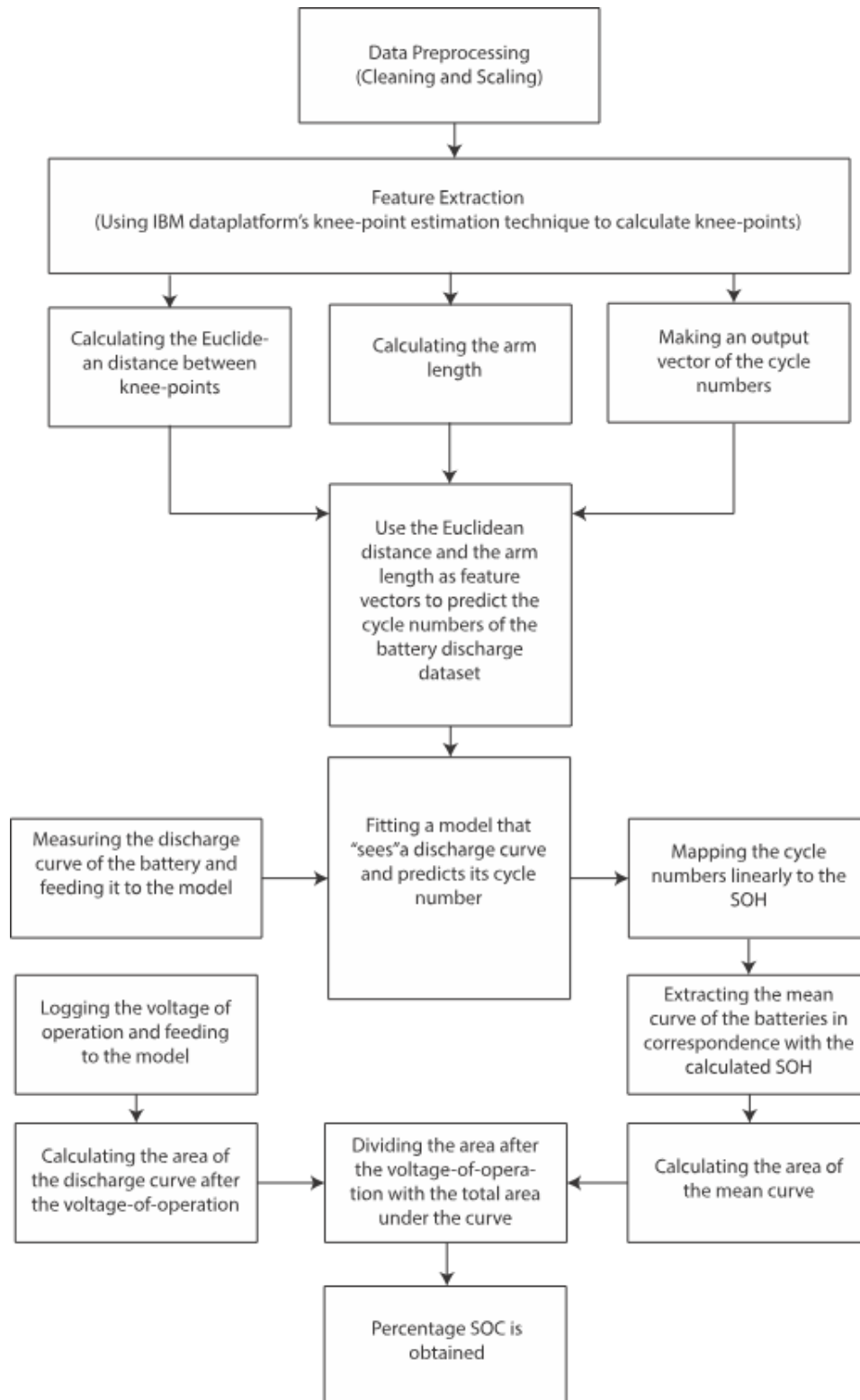


Figure 4.2: Block diagram of SOH and SOC estimation with machine learning.

[42] for the knee point estimation. More specifically, the calculation of the knee points involves two steps; first we calculate the equation of a line from the highest voltage  $V_{max}$  to the lowest voltage  $V_{min}$  of the curve, i.e., a straight line passing through points A and B. We then calculate the largest perpendicular distance from the line  $AB$  to the discharge curve. The longest perpendicular distance will represent one of the knee-points of the curve denoted as D.

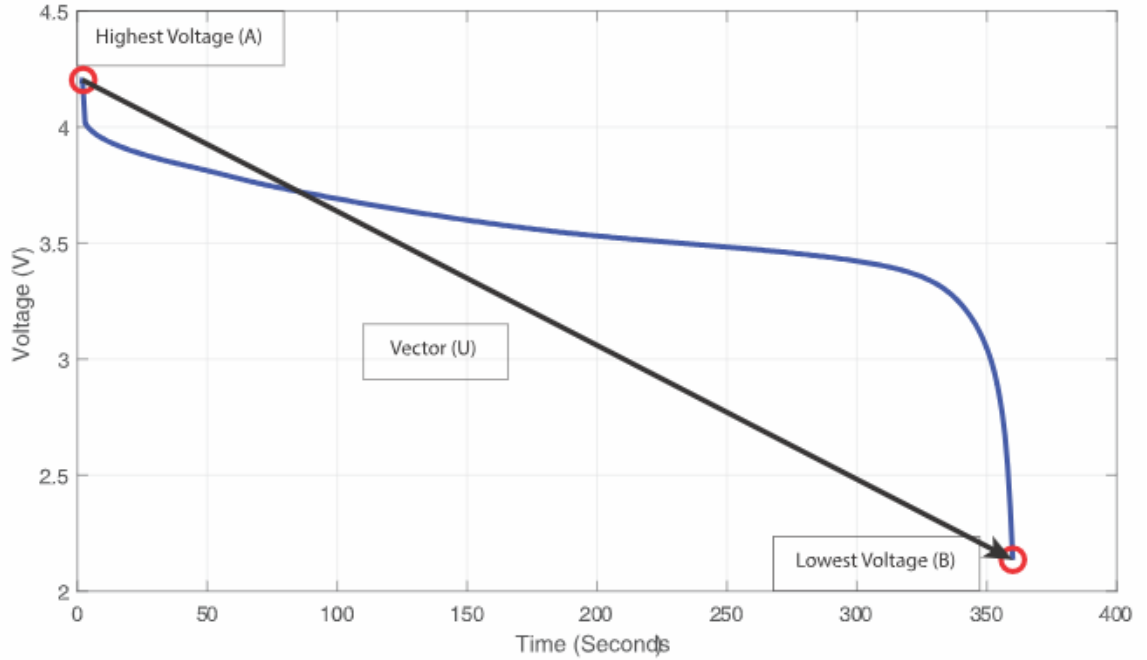


Figure 4.3: An imaginary line for Knee points calculation.

The second knee point, C, can be calculated with a similar approach by applying the algorithm in a region between the start of the curve and the point of intersection between the graph and the determined line  $AB$ .

To calculate the perpendicular distance from the line to the graph, we use the dot product of vector algebra. Let  $AB = V$  and let there any  $N$  points sampled from the discharge curve. Note that the data set already contains  $N$  points for each curve of battery discharge. Let  $N = \{n_1, \dots, n_N\}$  be the set of all points on the curve such that  $|N| = N$  where  $| \cdot |$  denotes the cardinality of the set  $N$ . For each  $n_i, i \in \{1, 2, 3, \dots, N\}$ , a vector  $W_i$  is

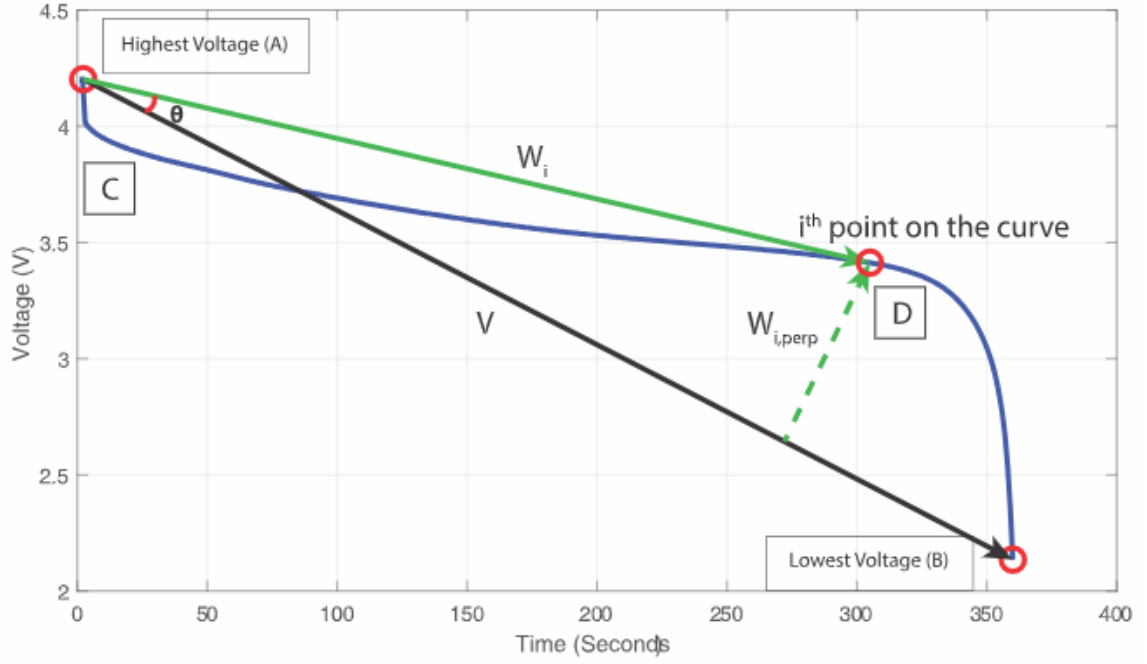


Figure 4.4: Calculation of 1st knee point

drawn from point  $A$  such that the set of vectors is denoted by  $W = \{W_1, \dots, W_i, \dots, W_N\}$ . For each  $W_i$ , a dot product between  $V$  and  $W_i$  is taken such that the angle between them is given as

$$\theta_i = \cos^{-1} \left( \frac{V \cdot W_i}{V W_i} \right), \quad (4.1)$$

where  $\cdot$  denotes the magnitude operator. The value of the perpendicular distance in terms of the single  $V, W_i$  pair is calculated by,

$$W_{i,perp} = W_i \cos \theta_i. \quad (4.2)$$

The knee-point  $D$ , is then found by taking the maxima, i.e.,

$$D = \max_i(W_{1,perp}, W_{2,perp}, \dots, W_{N,perp}) \quad (4.3)$$

For calculating the other knee-point,  $C$ , the range of the search has to be reduced. This

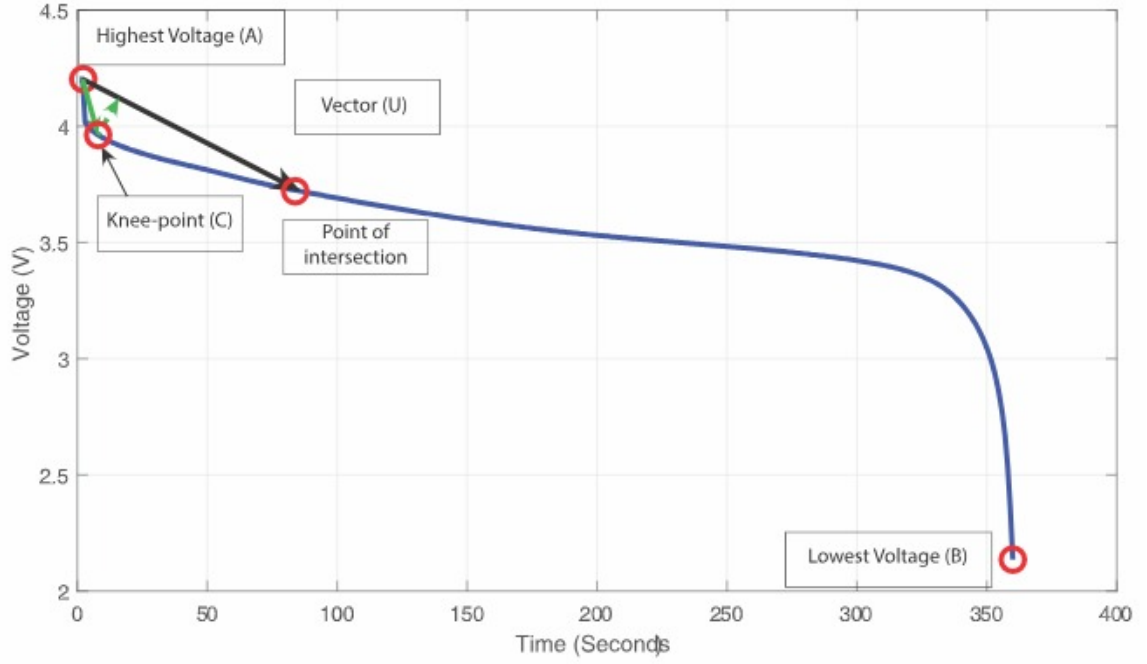


Figure 4.5: Calculation of 2nd knee point

is done by finding the intersection of the line  $AB$  and the discharge curve and setting the upper limit of the search equal to the intersection point. Thus  $N$  reduces here to a lower value.

## 4.2 Feature Engineering

An important step in machine learning is the preparation of the dataset to fit our needs. After logging the discharge voltage and time for points A, B, C and D of the battery dataset, we extract features that would be valuable input parameters in our machine learning model. In order to extract features for the training of our machine learning model, we calculate the Euclidean distances  $d_{AC}$ ,  $d_{CD}$  and  $d_{DB}$ . It is to be noted that the relative health of the battery decreases with a decrease in the distance between the two knee-points. Hence, our first input parameter becomes the Euclidean distance between the two knee-points of the discharge curve, i.e.,  $d_{CD}$ . The line between points C and D is termed the pseudo-linear region due to its linear nature. Now that we have our first input parameter in the form of

the pseudo-linear region, we now analyze the lengths  $AC$  and  $DB$  as other input parameters to the ML model. The input parameters in ML algorithms are chosen to be linearly independent to cater for redundancy. As we set our first feature as the Euclidean distance  $d_{CD}$ , we need to know what other features are linearly independent to our first feature. Fig. 6 represents the relationship between  $d_{CD}$  and the other two candidate features, i.e.,  $d_{AC}$  and  $d_{DB}$  for one of the batteries. As  $d_{DB}$  provides a more non-linear relationship to  $d_{CD}$  than  $d_{AC}$ , we therefore, select our second input feature to be  $d_{BD}$ . We call the length  $d_{CD}$  as *arm length*, as it is the length of the offshoot discharge arm of the battery voltage curve.

### 4.3 Data Preprocessing

The features extracted from each of the discharge curves is used to create the input data matrix, as

$$X = [x_1, x_2] \quad (4.4)$$

where  $x_1$  and  $x_2$  are the  $(D \times 1)$  vectors representing the extracted features from the discharge curves if  $D$  is the number of discharge curves of a battery. Our data set contains the discharge curves of a battery in a sequential order. The first discharge curve is assumed to correspond to a healthy battery. After one cycle, the next curve is obtained by charging and discharging this battery again. Hence the  $n^{th}$  discharge curve is taken on a battery that has undergone  $n$  charges and  $(n - 1)$  discharges; consequently losing some of its health. We use this characteristic of the data set to create the output vector of the machine learning model. The final form of the data set is;

$$\Delta = [X, y]. \quad (4.5)$$

where the each element of  $y$  denotes the cycle number corresponding to  $x_1, x_2$  features.

#### 4.4 Machine Learning for SOH

This data set is fed into machine learning models to determine the best fit. The root mean squared error (RMSE) of each model is calculated to analyze the authenticity and feasibility of the model. We have concluded that a polynomial regression model of  $3^{rd}$  degree provides the best fit with the least RMSE. The hypothesis equation utilized for regression is given below

$$h_{\theta}(x_1, x_2) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_1 x_2 + \theta_5 x_1^3 + \theta_6 x_1^2 x_2. \quad (4.6)$$

The values of the parameters are determined by using the gradient descent algorithm. This algorithm quantitatively compares the predicted value of cycle number from the model to the ground truth  $y$  of the data set. The following equation calculates the RMSE of the machine learning model;

$$J(\theta) = J(\theta_0, \dots, \theta_6) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x_1^{(i)}, x_2^{(i)}) - y_m)^2. \quad (4.7)$$

where  $J$  is the cost or error function quantifying RMSE of the model,  $m$  is the size of the data set and  $\theta$  is the vector containing model parameters  $\theta_0, \dots, \theta_6$ . To minimize this error, we must tweak the model parameters and find the optimum values recursively. To get the new values of the model parameters  $\theta_{n+1}$ , we take the partial derivatives of the error term with respect to each of the model parameters in vector  $\theta_n$  and subtract this derivative term from subtracted from  $\theta_n$ .

$$\theta_{n+1} := \theta_n - \alpha \frac{\partial}{\partial \theta_n} J(\theta_0, \dots, \theta_6). \quad (4.8)$$

The process is then repeated with the new parameter values to minimize the error function and push it closer to zero. An ideal cost function will have a value of 0.

The first discharge cycle will correspond to a battery health of 100% and the last dis-

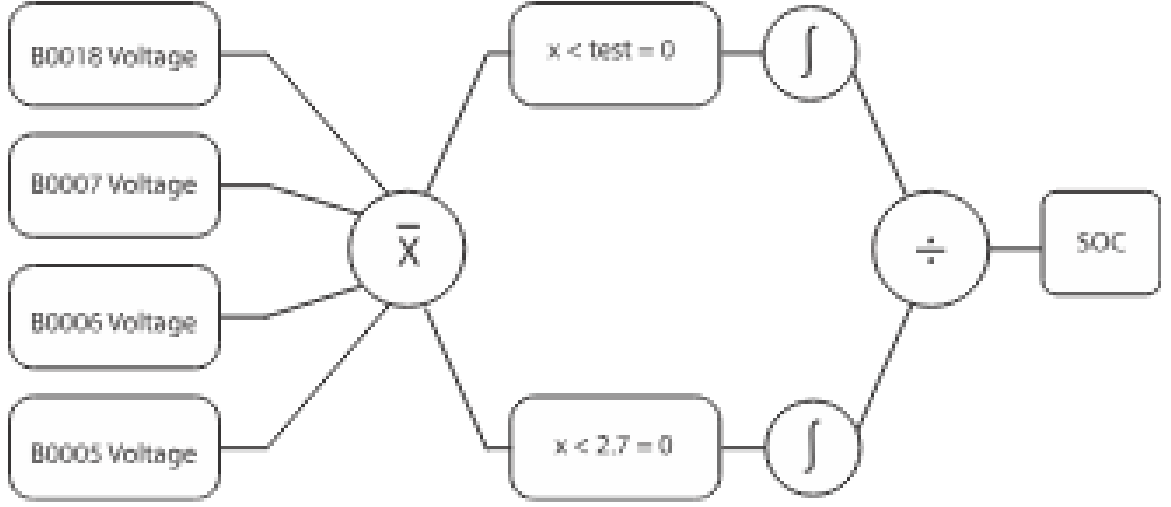


Figure 4.6: Algorithm for estimation of SOC.

charge cycle will provide the discharge curve of the least "healthy" battery.

#### 4.5 SOC estimation

The machine learning model uses the entire discharge curve to estimate the SOH of the battery. Now we provide another parameter to the algorithm; the current voltage-of-operation. Our goal is to provide an estimated amount of charge, that is left in the battery at a certain voltage-of-operation ( $V_o$ ). To do so, we use the curve from the maximum  $V_o$  to the minimum  $V_{min}$ . As provided in the NASA Ames prognostic repository, the batteries have been discharged till +2.7V, hence we will only use the curve upto that point.

After extracting the voltage curve from  $V_{max}$  to +2.7V, we calculate the area under the curve,  $A_T$ . Now we use the given voltage  $V_o$ ; the current voltage-of-operation of the battery. The total charge remaining in a battery of a specific SOH, at a particular  $V_o$  is calculated by dividing the area under the curve after  $V_o$  to +2.7V, by the total area of the curve. This gives an estimate of the percentage of charge left in the battery at a particular voltage level. The formula for  $SOC\%$  calculation by

$$SOC\% = \frac{\int_{V_o}^{+2.7V} V(t)dt}{A_T}, \quad (4.9)$$

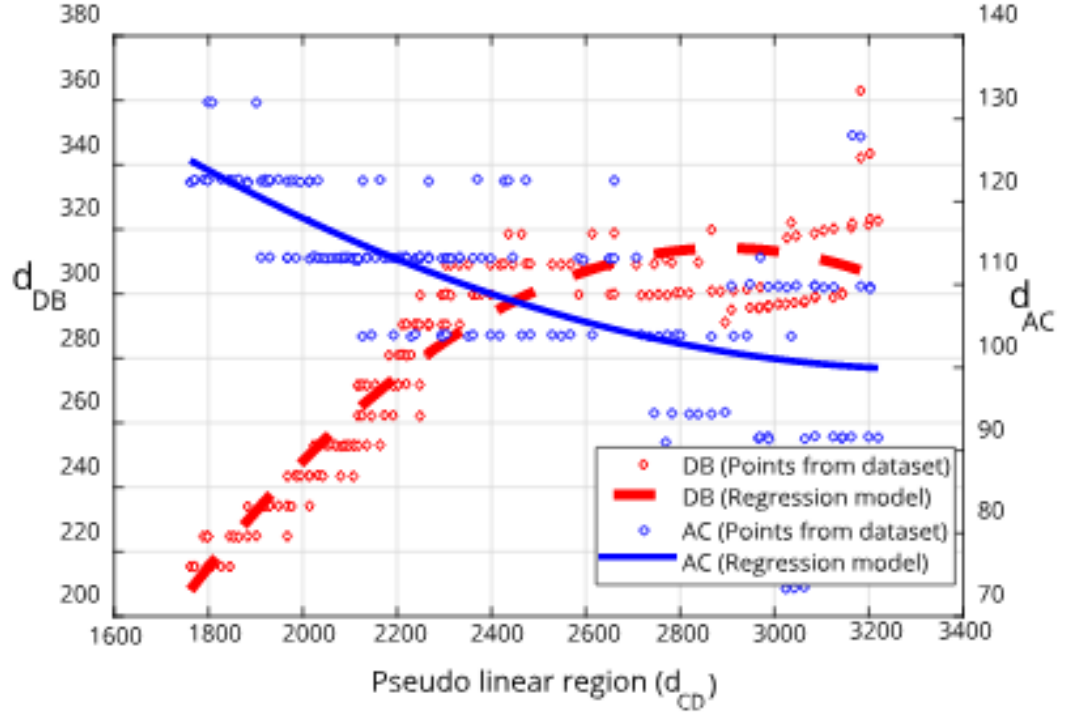


Figure 4.7: Relationship between  $d_{AC}$  and other two features for battery.

where  $V(t)$  is the discharge curve function with respect to time and the nominator term represents the area from  $V_o$  to  $+2.7V$ .

## **4.6 Summary**

This chapter summarized many main concepts purposed of this thesis including, machine learning technique for state of health (SOH) and also purposed method of state of charge (SOC) relationship with SOH. Moreover, this chapter has provided details what important steps are required to implement machine learning, such steps are data labeling, data processing, feature selections and algorithm purposed for selection of such features with the help of knee-point calculation method.

## CHAPTER 5

### RESULTS AND ANALYSIS

#### 5.1 Results of SOH

We have implemented machine learning on six batteries with different current and temperatures levels respectively, i.e, first three batteries with the current level of 1A and temperature of  $24^{\circ}C$ , similarly, the second three batteries with the current level of 2A and temperature of  $4^{\circ}C$  are modeled. it is to be noted each battery has different cycles, therefore, each battery represents a different trend when plotted against the pseudo-linear region ( $d_{CD}$ ) and arm length ( $d_{DB}$ ).

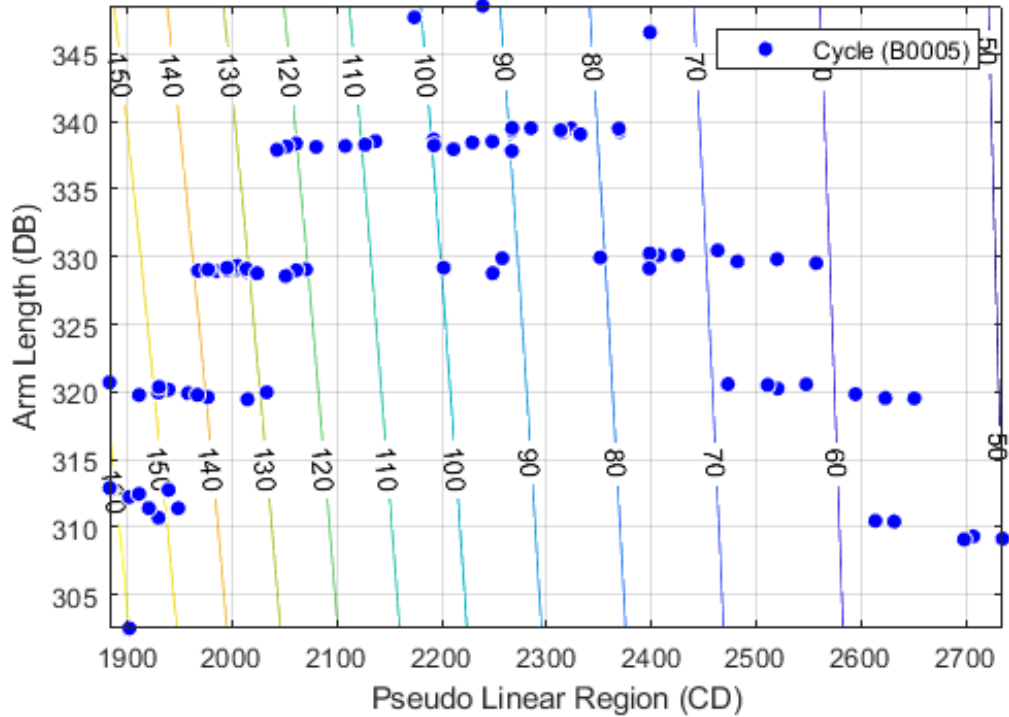


Figure 5.1: Contour plot of Battery 1 at  $24^{\circ}C$

Fig.5.1, 5.6, and 5.9 represent the contour plots of the models used for the estimation of

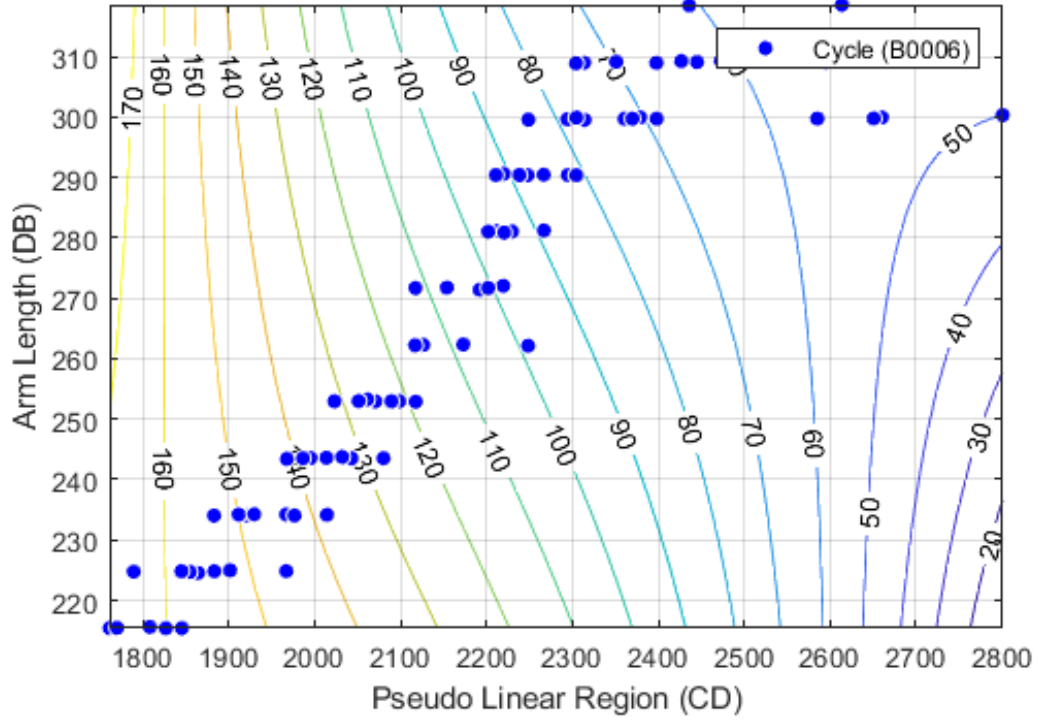


Figure 5.2: Contour plot of Battery 2 at  $24^{\circ}C$

the SOH of the batteries at  $24^{\circ}C$ . The training dataset used in each case is shown by points on the graph. It can be inferred from the graph that with the increase in the  $d_{CD}$  and  $d_{DB}$  the SOH also increases. It should be noted that the SOH is given in the form of a percentage metric but the contour lines go above 100% on the right side of the contour plots. The reason behind this apparent anomaly is that the machine learning model predicts the SOH without any regard to the nature of the SOH. For our model, SOH is just a numeric value though there are no "real" data points in the above 100% region. Similarly, The Fig.5.10, 5.11, and 5.12 show contour plots of the models estimating the SOH at  $4^{\circ}C$ . As evident from the plots, the SOH at  $4^{\circ}C$  for the same value of  $d_{CD}$  is better than that for a battery at  $24^{\circ}C$ . in contour plot of each of battery results it is to be noted that the blue circles show the cycles, whereas the lines correspond to SOH, similarly if these points are connected together a we get a  $3^{rd}$  polynomial trend shown as represented by equation 10.

It can be inferred from the graph that with the increase in pseudo-linear region ( $d_{CD}$ )

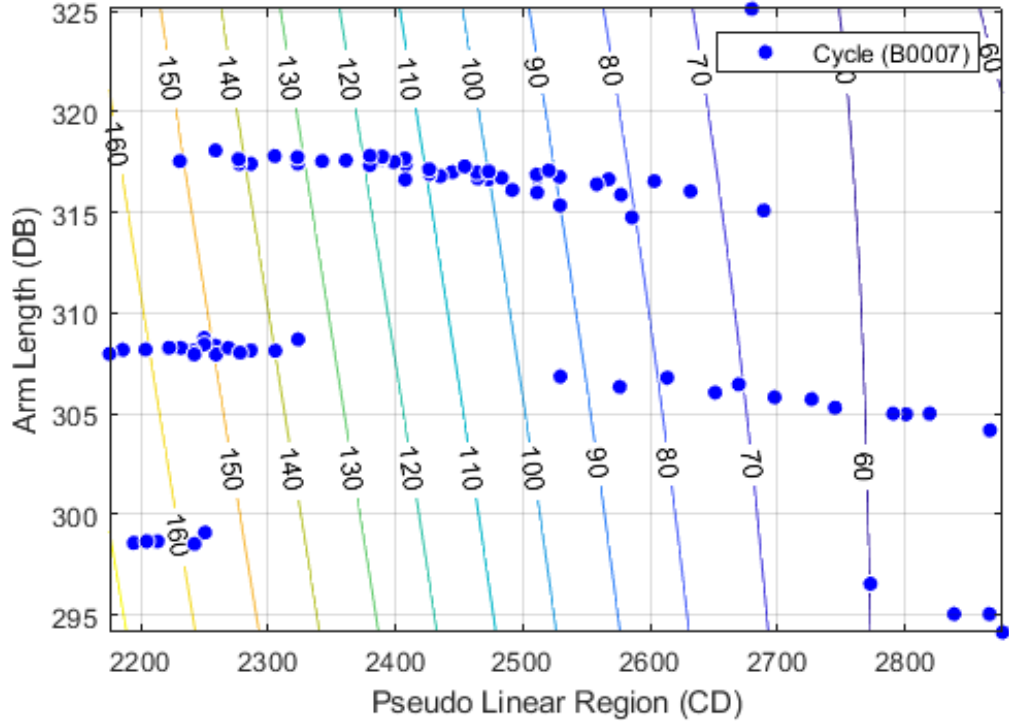


Figure 5.3: Contour plot of Battery 3 at 24°C

and arm length ( $d_{DB}$ ) the SOH also increases. It should be noted that the SOH is given in the form of a percentage metric but the contour lines go above 100% on the right side of the contour plots. The reason behind this apparent anomaly is that the machine learning model predicts the SOH without any regard to the nature of the SOH. For our model, SOH is just a numeric value though there are no "real" data points in the above 100% region.

## 5.2 Result of SOC

The discharge curve labeling was one of the most important tasks which has been implemented by knee point calculation. This calculation not only accurately scooped out the threshold voltages for the feature estimation but also provides freedom of time and again assumptions require for the knee point calculation of each discharge curve individually. We have extracted  $d_{CD}$  and  $d_{DB}$  as useful features as  $d_{CD}$  represents the capacity of the battery and  $d_{DB}$  provides remaining 10% of charge with EOD point which not only helps

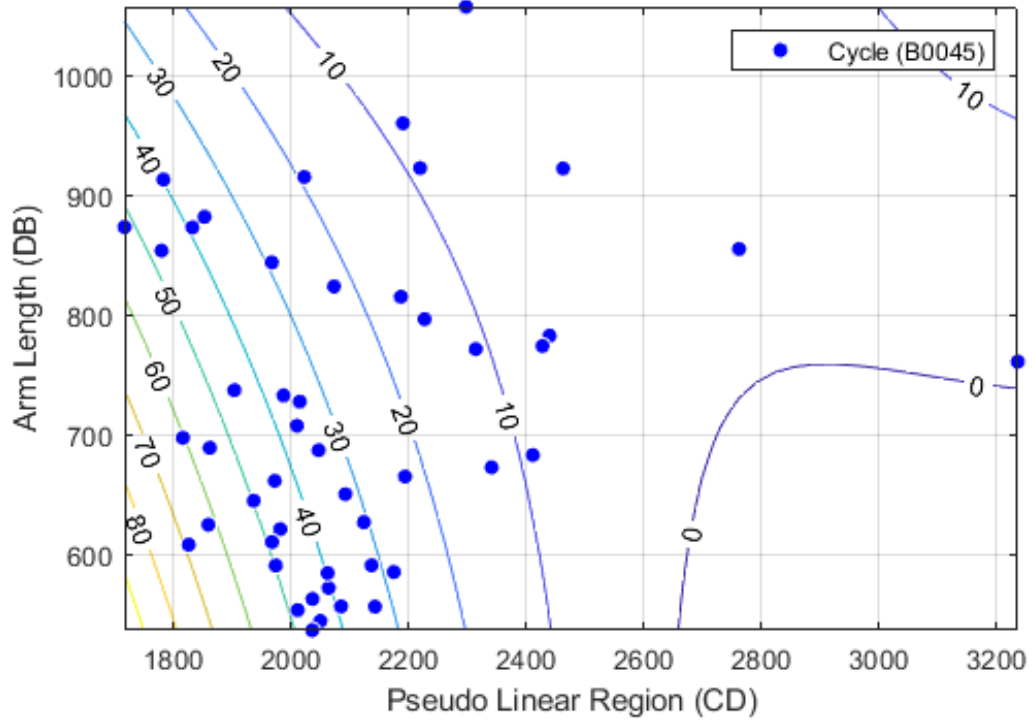


Figure 5.4: Contour plot of Battery 1 at  $4^{\circ}C$

in finding SOH but also for SOC at any instance of time. Moreover, due to lack of data set, this algorithm could have also estimated SOH of batteries at higher temperature and load current levels, nevertheless, implementing the purposed algorithm using  $4^{\circ}C$  and  $24^{\circ}C$  temperature and 1A and 2A currently levels, respectively are sufficient verification of this method.

Fig.5.13 shows the variation of the SOC at a particular voltage with the SOH of the battery. This variation shows a decreasing trend because the increase in the SOH of the battery causes the overall area of the curve to increase. This increased area is in the denominator of the term that calculates the percentage SOC and henceforth, decreases the SOC. This percentage decreases much more rapidly as compared to a healthy battery because of its reduced capacity. Thus, it can be concluded from the graph that, for a particular value of the voltage, the SOC of the battery increases with the decrease in health, i.e., the battery loses charge faster when it has a low SOH. While observing Fig.5.13 the overall curve is uneven

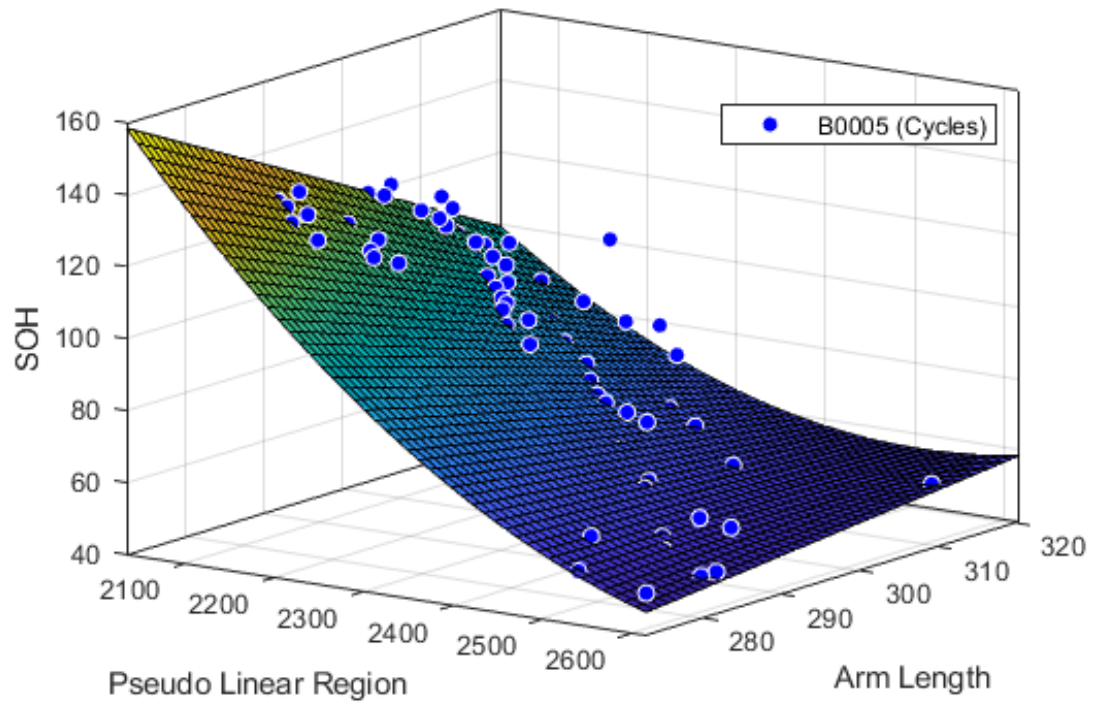


Figure 5.5: 3D plot of Battery 1 at 24°C

and have higher peaks at the ending, which should be low corresponding to time, the main reason for such abnormality is due sparsity of data set. The inclusion of such result may depict rough estimation, however, it is very important to draw a relationship between SOH and SOC together for a particular voltage which not only broadens the understanding of these two parameters but also introduces a correlation between them.

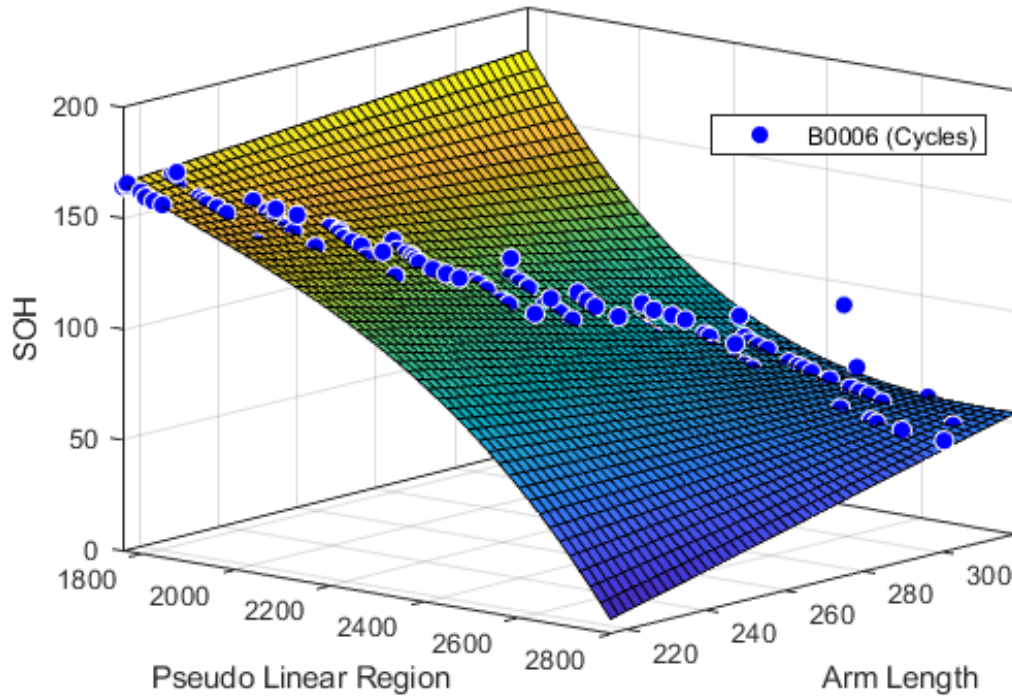


Figure 5.6: 3D plot of Battery 2 at 24°C

### 5.3 Accuracy of The Model

The data sets of batteries are divided into two major portions; training and testing. Such classification of data is important to verify the performance of the developed models. In doing so, our classification is purely done randomly or in other words, out of 100% of discharge curves, 70% of them are randomly chosen for training and remaining 30% of the discharge curves are dedicated for testing of the model.

First, we present our results on the relationship between the two extracted features of a battery curve versus the cycle number or SOH. Note that the data set contains the cycle number of each discharge curve, whereas the battery health is generally taken in terms of SOH. Therefore, we performed a mapping between cycle number and SOH such that greater the cycle number, the lesser is the SOH.

Finally, its worthwhile noting that accuracy of batteries (B0045-B0047) with temper-

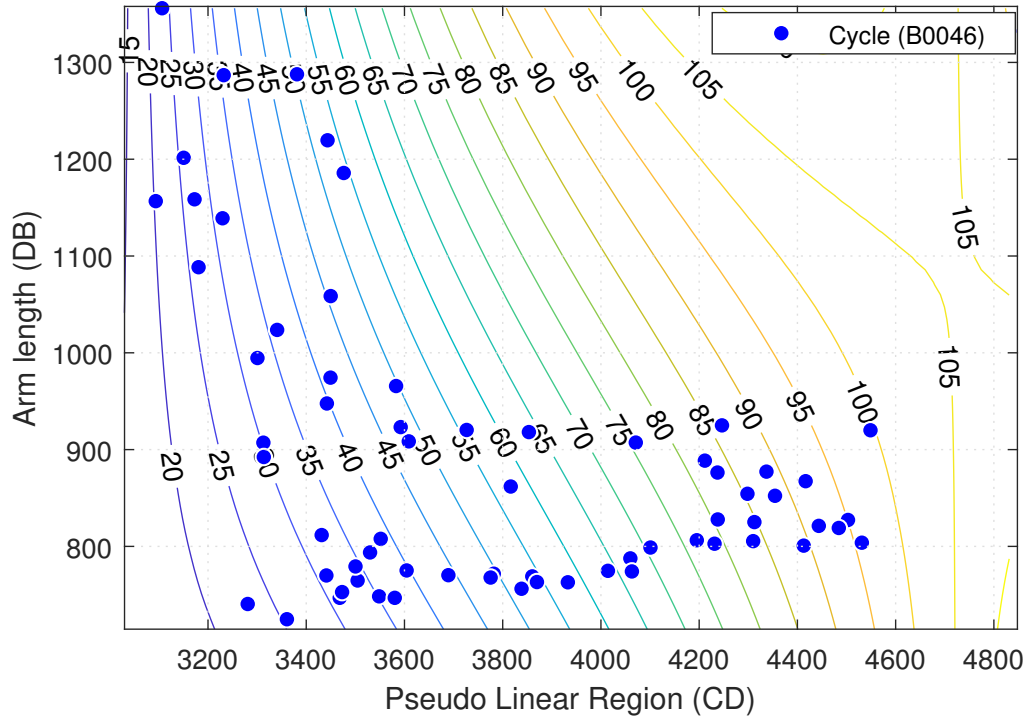


Figure 5.7: Contour plot of Battery 2 at  $4^{\circ}C$

Table 5.1: SOH Results Using Machine Learning Model for Different Batteries and Conditions

| Battery | Error (rms) |
|---------|-------------|
| B0005   | 4.1308      |
| B0006   | 4.7078      |
| B0007   | 7.0456      |
| B0045   | 8.1308      |
| B0046   | 12.1196     |
| B00047  | 14.7087     |

ature  $4^{\circ}C$  and 2A has lower accuracy (87.86% to 77.87%) in comparison to that of  $24^{\circ}C$  and 1A batteries accuracy (98% to 93.08%). this is due to the reason, the data set was very small and there excite larger gaps between adjacent discharge curves which increases error between training versus testing results. Such an error can be minimized with larger data sets. The accuracy as shown in Fig.5.14 is calculated by the following equation:

The Fig.5.10, 5.11, and 5.12 show similar contour plots of the models estimating the

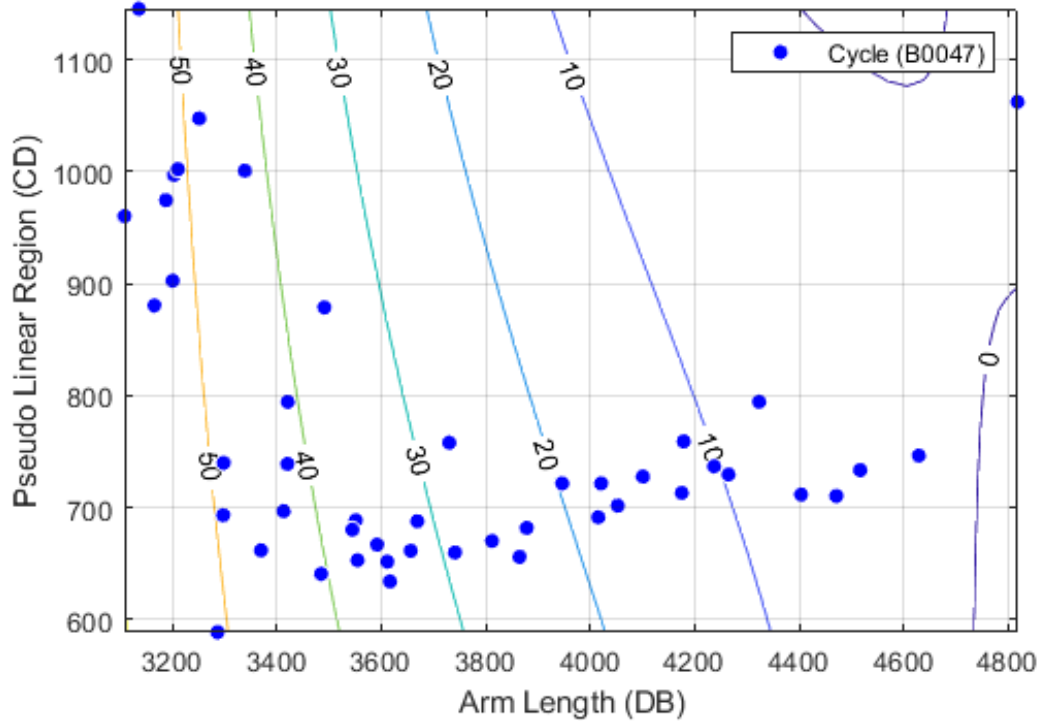


Figure 5.8: Contour plot of Battery 3 at 4°C

SOH at 4C. As evident from the plots, the SOH at 4C for the same value of pseudo-linear region ( $d_{CD}$ ) is better than that for a battery at 24C.

lastpage

$$Accuracy = 100 - \left[ \sum_{i=1}^{N_{test}} \frac{S\hat{O}H_i - SOH_i^{(t)}}{SOH_i^{(t)}} \right]$$

(5.1)

whereas,  $N_{test}$  are the number of discharge cycles of test data set,  $S\hat{O}H_i$  is the  $i_{th}$  term of estimated SOH and  $SOH_i^{(t)}$  is the  $i_{th}$  true value of discharge curve driven by training data set.

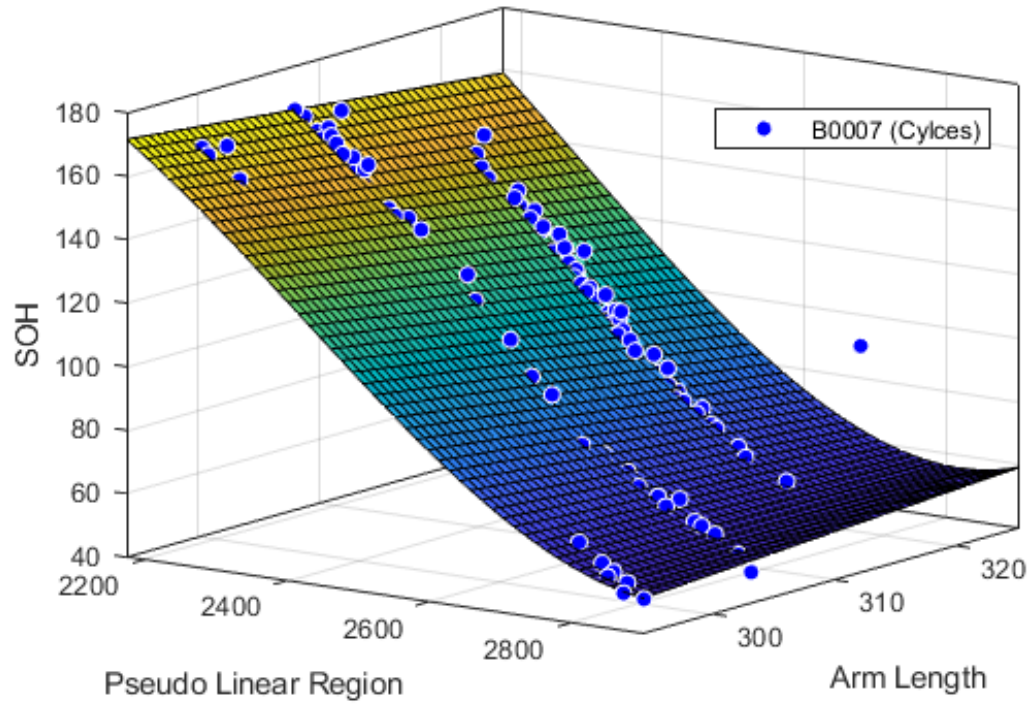


Figure 5.9: 3D plot of Battery 3 at 24°C

## 5.4 Summary

**This Chapter**, contains all results related to SOH of all different batteries such as batteries with different current and temperature levels respectively. This chapter also provided detail how such results are drawn. and finally, SOC versus SOH result displays an important relationship between charge of battery and degradation process and, provided discussion on the drawn results and also shares how such results are important towards contribution in estimation of Li-ion batteries, while keeping a balance between two different domain knowledge, i.e., power engineering and machine learning.

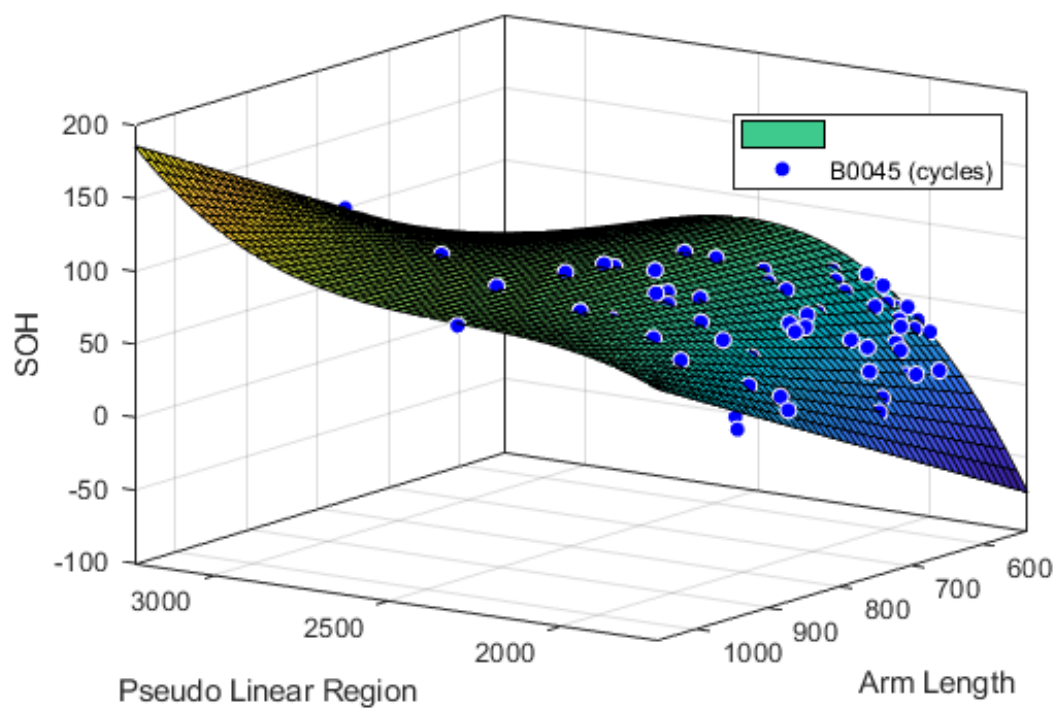


Figure 5.10: 3D plot of Battery 1 at 4°C

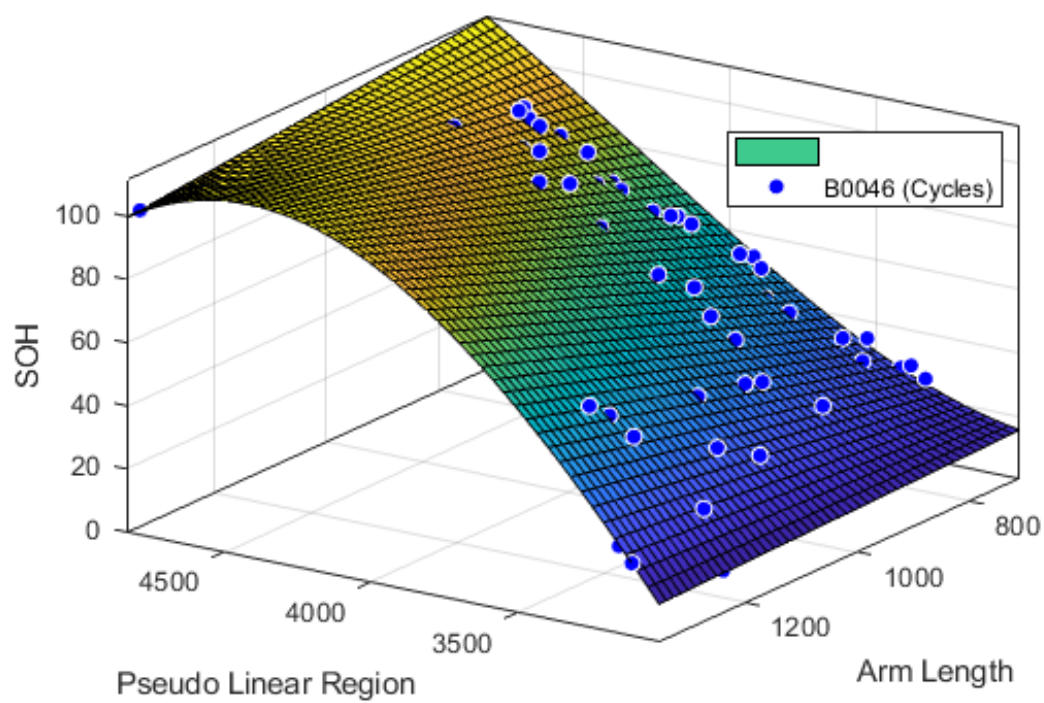


Figure 5.11: 3D plot of Battery 2 at 4°C

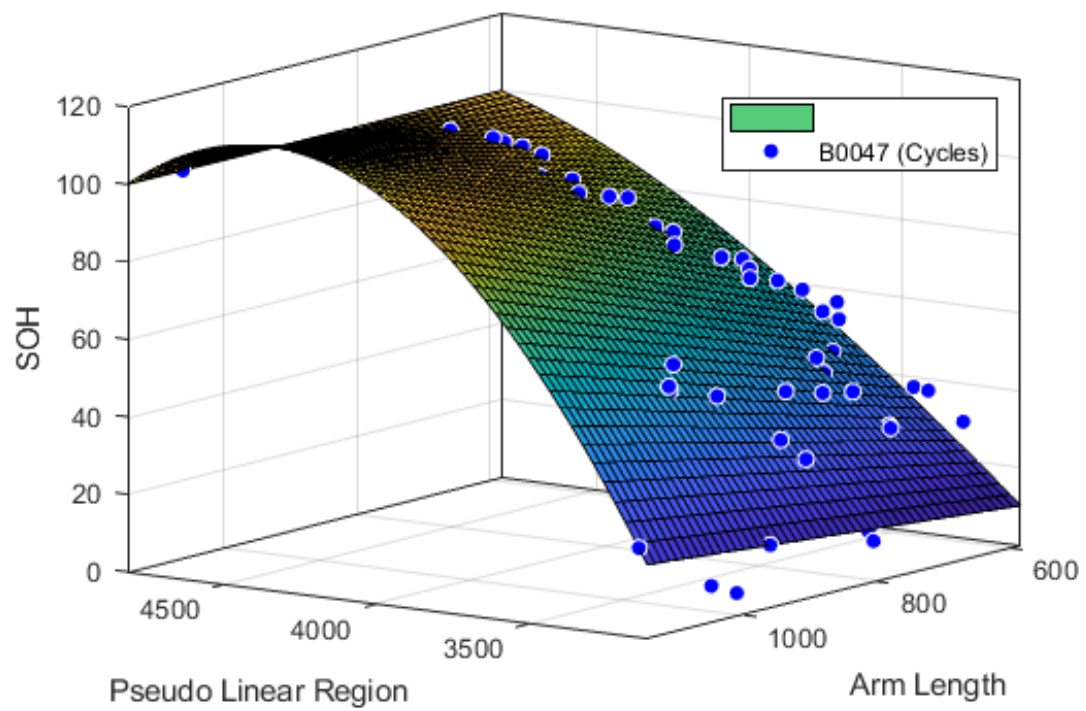


Figure 5.12: 3D plot of Battery 3 at 4°C

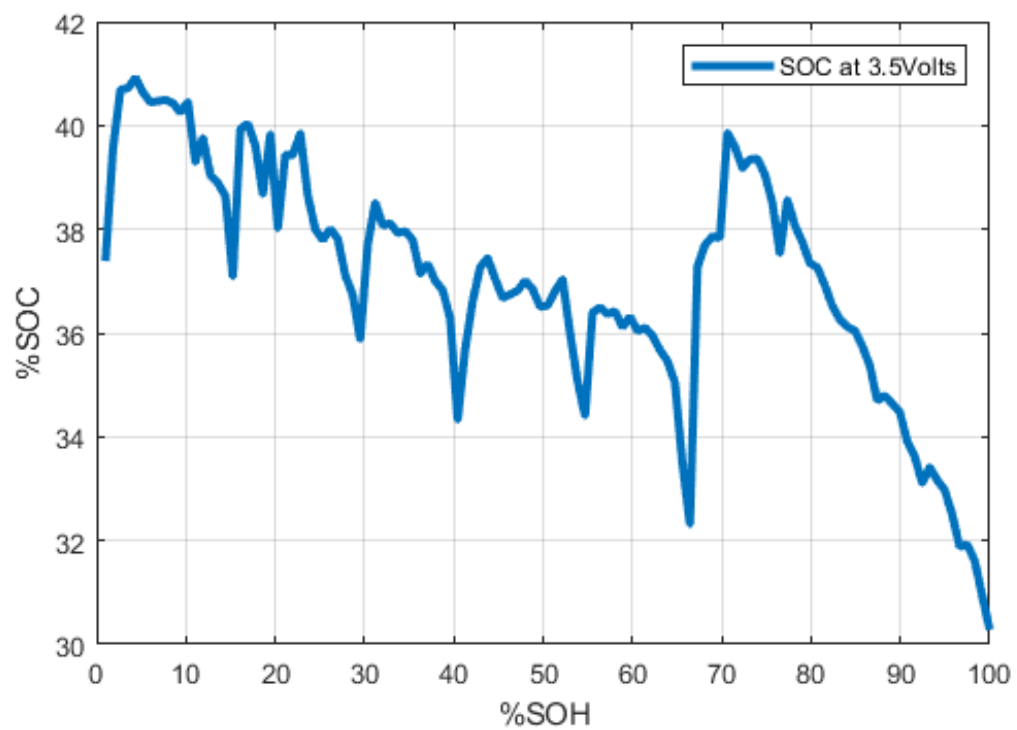


Figure 5.13: Relationship between SOC of a particular voltage and SOH

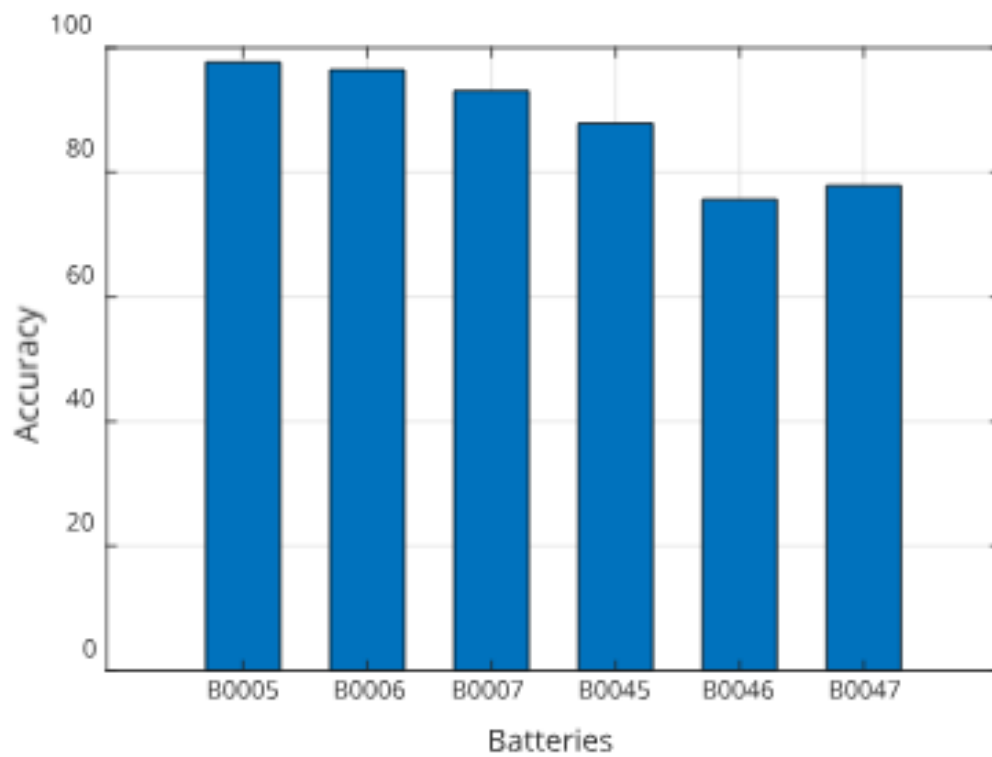


Figure 5.14: Accuracy results of training versus testing datasets of batteries

## **CHAPTER 6**

### **CONCLUSION AND FUTURE RECOMMENDATIONS**

This work summarized some of the important contributions for battery management systems (BMS) techniques. while the BMS is consist of different parameters to ensure battery operates well with respect to electrical parameters, chemical properties and also have temperature balance during its normal and abnormal conditions such as over charging and discharging or drawing current beyond the capacity of battery which causes permanent damages to its internal structure.

In initial stage of this research, important findings are made in the estimation of SOH and SOC using the machine learning approach. As the behavior of the battery is non-linear and can not be predicted directly, therefore, an indirect estimation and prediction method was needed such as machine learning. In estimation of battery SOH and SOC, first extracted features from discharge curves of different batteries with current and temperature levels, i.e., B0006 is a battery with 2A and  $24^{\circ}C$  has been done, whereas, B0047 is a battery with 1A and  $4^{\circ}C$ . During the extraction of features, a knee-point calculation technique has been implemented which identified knee-points numerically.

The features are fed to our machine learning models, which provide SOH over all discharge cycles of the battery with respect to the pseudo-linear region and arm length. In addition, SOC% is estimated by considering the area under the curve of the measured voltage. This technique has not only predicted correct SOC% for a discharge cycle but has also provided degrading trend with respect to all training cycles. In other words, if we are given a battery at any instant of time, SOC can be estimated by considering voltage as a reference point and using reverse engineering to calculate the SOH. The future works can be carried out with the help of this research work; Due to a limited dataset, the error was large which could be minimized with a larger dataset. the Polynomial regression models

stand out over other machine learning techniques as they provide the best RMSE. However, with large datasets, better methods can be implemented, e.g., artificial neural networks and support vector machines and finally our models need the entire discharge curve to calculate the SOH and SOC of the battery. Provided a larger dataset, Recurrent neural networks can be used to do a online estimation of SOH.

# **Appendices**

## APPENDIX A

### ADDITIONAL WORK

#### A.1 Battery impedance characterization using Arduino

The noise has been used for different applications, e.g., for the identification and recognition of system behavior in mathematics, communication, control, and even power engineering [1],[2], i.e., in our case for extraction of battery parameters [3]. Several methods are available for the generation of noise signals, however, this introduces system level complications, e.g., requires extra hardware for large memory space, slow processing speed and cannot be reproducible when it is required to be regenerated for data logging [4].

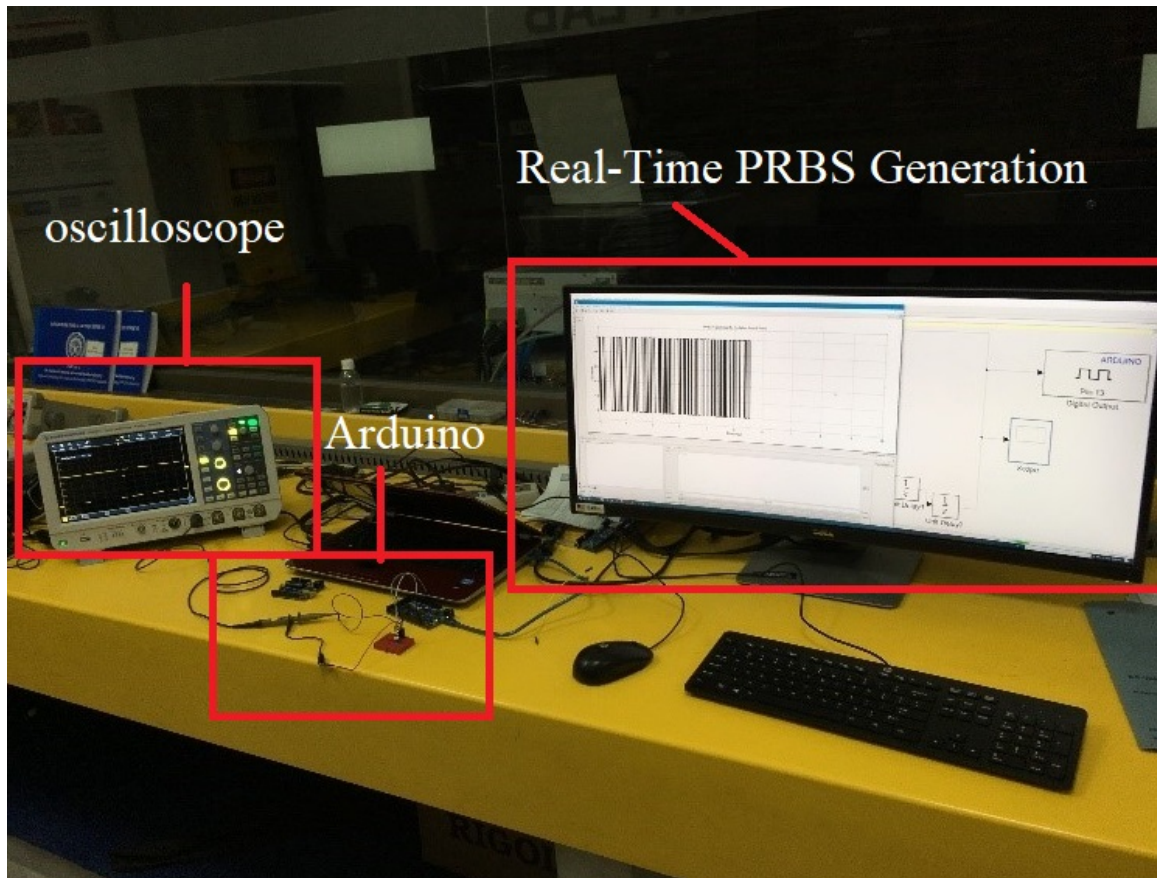


Figure A.1: Hardware: Arduino board, Matlab interfacing and oscilloscope.

In this scenario, Pseudo-random sequences provide enormous support. Pseudo-random sequences do not need large memory as they can generate a sequence with only a seed as input random number and they keep on generating a sequence with the help of multiplying the seed value i.e., middle square algorithms or Mersenne twister Method: Linear Congruential Generator (Modulus, multiplier, and increment) [5] however, in PRBS after definite time interval signal repeats itself due to lack of combinations (possible combination in middle square algorithm can be 9,999 and in case of Linear Congruential Generator has a larger combination in terms of  $2^n$  where  $n$  is an integer) and finally have a periodic behavior after the some interval [6]. In MATLAB/Simulink we have generated a sequence with delay and logic (XOR) function blocks respectively. In doing so the number of delay and Logic gates create different bit size PRBS however in our research we have generated 8-bit PRBS for future findings. This paper is intended to provide a real-time integration of Arduino microcontroller with MATLAB/Simulink. Recently, Arduino boards [7] have emerged as the most user-friendly and rampantly demanding microcontrollers for general purpose applications for engineers, hobbyists, medical testing, agriculture, industry, academia and even for researchers with very less knowledge of electronics [8-12]. Here Arduino is used for two applications; first for the PRBS signal generator as a function generator, and second, as monitor as a Low-cost replacement of oscilloscope. Moreover, this research paper provides a comparison of the two eminent Arduino board family (i.e., Arduino Due and Arduino Uno) for consumer adoption. Lastly, we have shared four different cases in which Arduino can be integrated and fulfill desire tasks. A systematic representation of PRBS block diagram is shown in Fig. A.2.

When it comes to physical interfacing of hardware with software plethora of equipment unfit to meet experimental requirement in-terms of delay in operation or in their accuracy, however, Arduino provides great brace in between hardware and software interfacing with a millisecond to millisecond accuracy. If we compare Arduino with other available devices which are highly costly and limited to some list of experiments use only, diffi-

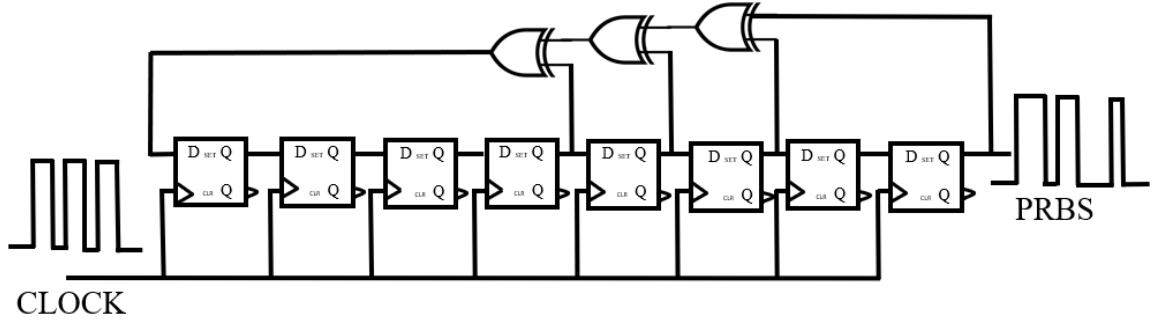


Figure A.2: Systematic representation of PRBS.

cult programming interface which requires good background knowledge of programming and electronics (i.e., National instruments I/O boards: DAQ Cards, Cambridge Electronics equipments, etc.), whereas Arduino standouts as multipurpose machine and provide great ease to experimenters with less background knowledge. In addition to this, many low-level experimental operations do not require specific software and costly boards, in our case, Arduino has replaced the function generators which are 1000s times expensive compared to an Arduino (an average advanced oscilloscope with PRBS function costs between 3,500 US Dollars whereas Arduino costs 3.5 US Dollars only). another salient feature of Arduino is it can be used as stand-alone applications (without the need of computer) and with a computer for monitoring (in this case as an oscilloscope) with parallely communicating with other devices, e.g., XBee and Bluetooth, etc.

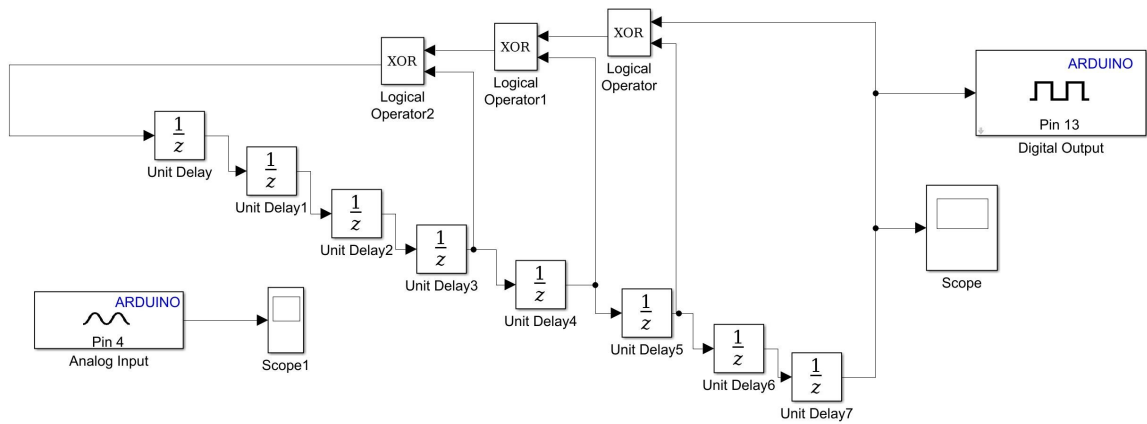


Figure A.3: System scale PRBS generation.

Arduino has open sources software package and hardware design with an Atmel AVR microprocessor. They are available preassembled and flexibility of variations for user-defined applications. Arduino has one of the largest community recordings to recent results from google survey estimated that roughly more than 14 million people search for Arduino and this number is rampantly increasing. By growing community Arduino has excellent online resource and add-ons for interfacing with different platforms like flash, MaxMSP, processing, and MATLAB/Simulink. These interfaces require a few lines of code or in some cases requires add-ons. Even after these resources, several videos on YouTube and blogs where a large community of students, researchers, and industrialist resolve problems. Third parties continuously releasing people required variations in the Arduino concept to directly solve problems by providing multipurpose Arduino shields for communication, DC motor controllers, displays and cameras, etc.

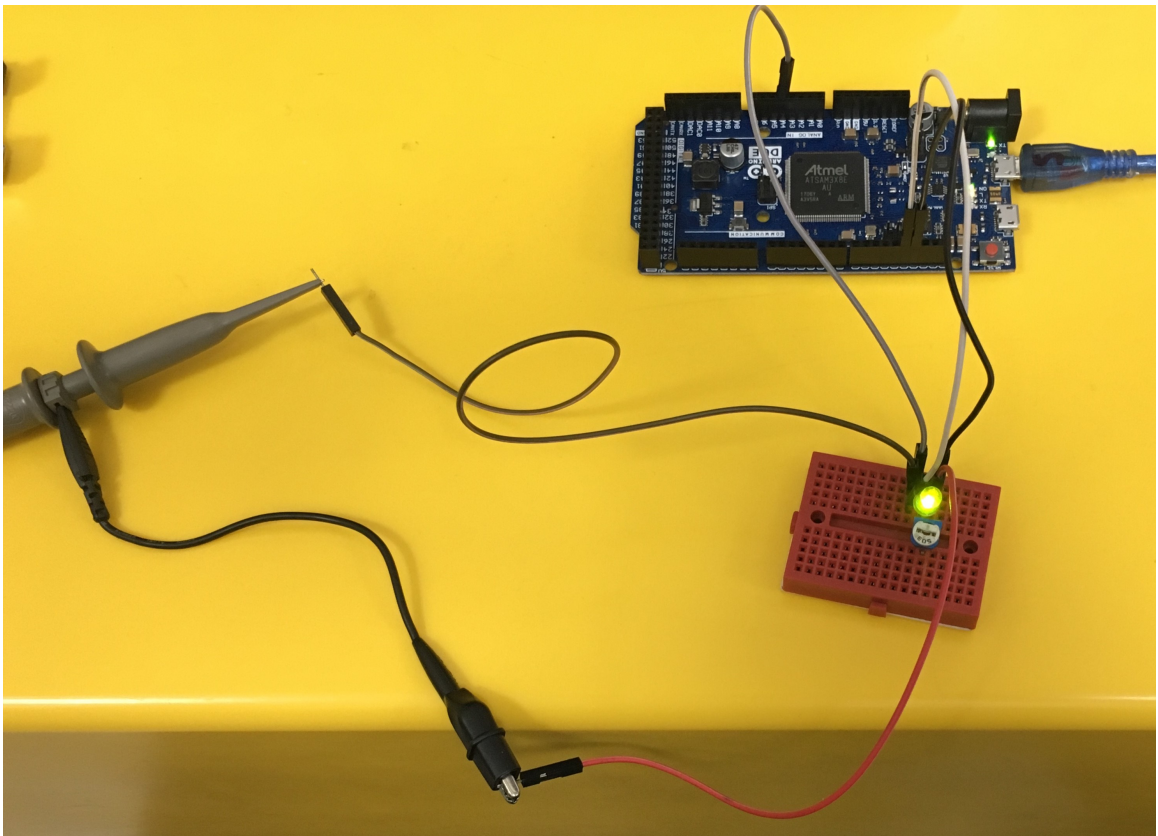


Figure A.4: Arduino connectivity with a LED for observation of PRBS.

PRBS signal and we have a different model of function generator with noise and no options for PRBS generation, therefore, we have designed Arduino base PRBS generation with the same accuracy and low-cost. Here we have also monitored our PRBS signal from Arduino which is the replacement of expensive oscilloscopes. four different cases to cross-validate results and accuracy of Arduino. In the first case, we observed PRBS signal by just visualizing indicator by Led light blinks. In second, the signal is observed on oscilloscope and analysis of different characteristics. In third, we designed another simulation setup for Arduino Uno as the output signal (oscilloscope) for our predesigned PRBS signal. This way we remove the need of oscilloscope and observed signals with the help of another low-cost Arduino. in lastly, we connected the output of PRBS to the same Arduino to itself for real-time observation.

The integration of Arduino with MATLAB/Simulink in following steps. MATLAB 2017 version is used for our simulation. In MATLAB it is easy to download and install new hardware by add-ons option. This way Arduino library can be integrated. We have integrated Arduino support package block (Support package block for Arduino hardware Common Digital Output Block) with our PRBS model. And added a scope for real-time observation of PRBS. The following flow diagram is shown which explains how different steps are taken to connect Arduino with MATLAB/Simulink.

#### A.1.1 PRBS Generation in Simulink

Pseudo-random binary sequence (PRBS) is generated using Simulink delay and logic gate block (XOR) respectively. In this paper, we have generated 8-bits binary sequence for our future research purpose which includes for the estimating natural response of battery and complex impedance extraction of batteries. PRBS can be generated for different bits length however it introduces several complications. moreover, PRBS frequency and different parameters can be optimized by using the following equations. A detailed work on PRBS is explained by Kehan Zhu et al [4]. For n-bit PRBS There can be N number of possible

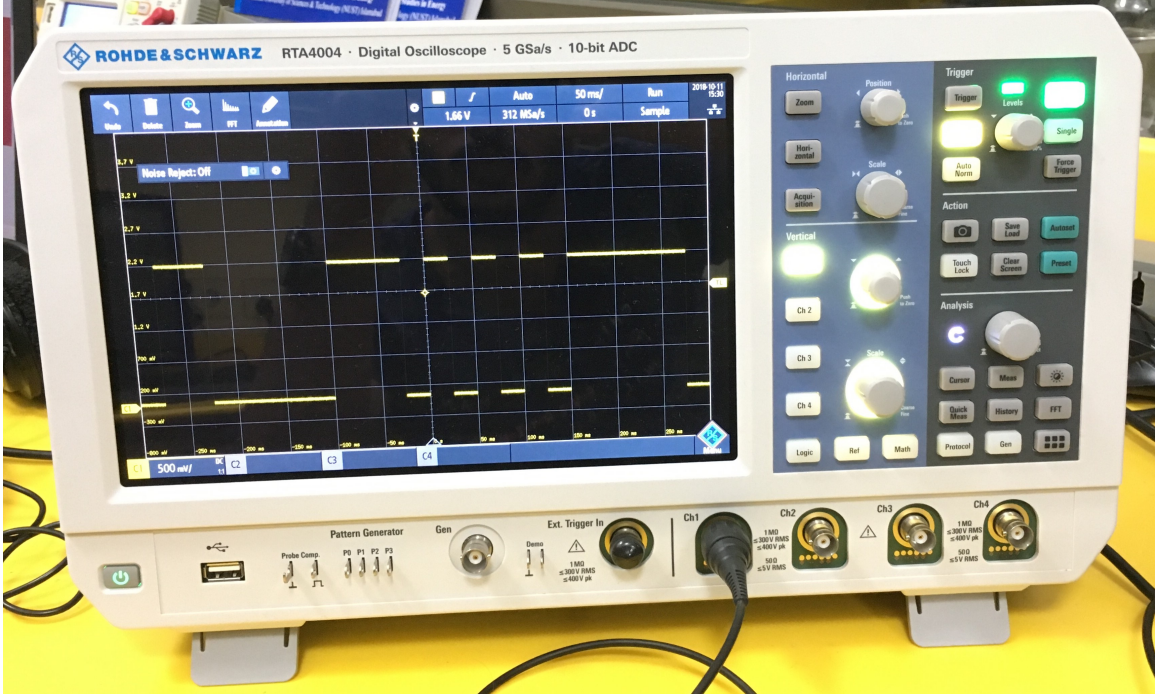


Figure A.5: Arduino based generated PRBS signal observation on oscilloscope.

states. In our case, 8-bit PRBS can have 255 different states. The PRBS that can have all possible states is called maximum length sequence. The bit length of the PRBS is incremental in nature, therefore, it takes time to run by every state. This way experimentation time will rise exponential over the increment of bits. Clock frequency ( $F_c$ ) is another imperative factor for selection of PRBS which can be optimized by changing delay time ( $T_c$ ) in delay function blocks respectively.

$$T_c = \frac{1}{F_c} \quad (\text{A.1})$$

A comparison between different PRBS bit length is shown in table below. which provides testing time and frequency clock frequencies are discussed for initial 1 periods, higher the clock frequency lowers the testing time and helps in covering a wider range of frequencies however this introduces analysis difficulties which are not the scope of this paper. The PRBS generation on Simulink is shown in Fig. A.7. here we provided 0.025 secs of sample time to each of delay function which generates a frequency of 40 Hz. Similarly, by

changing sample time we can achieve different frequencies for a given 8-bit PRBS. We have observed the output in four different ways.

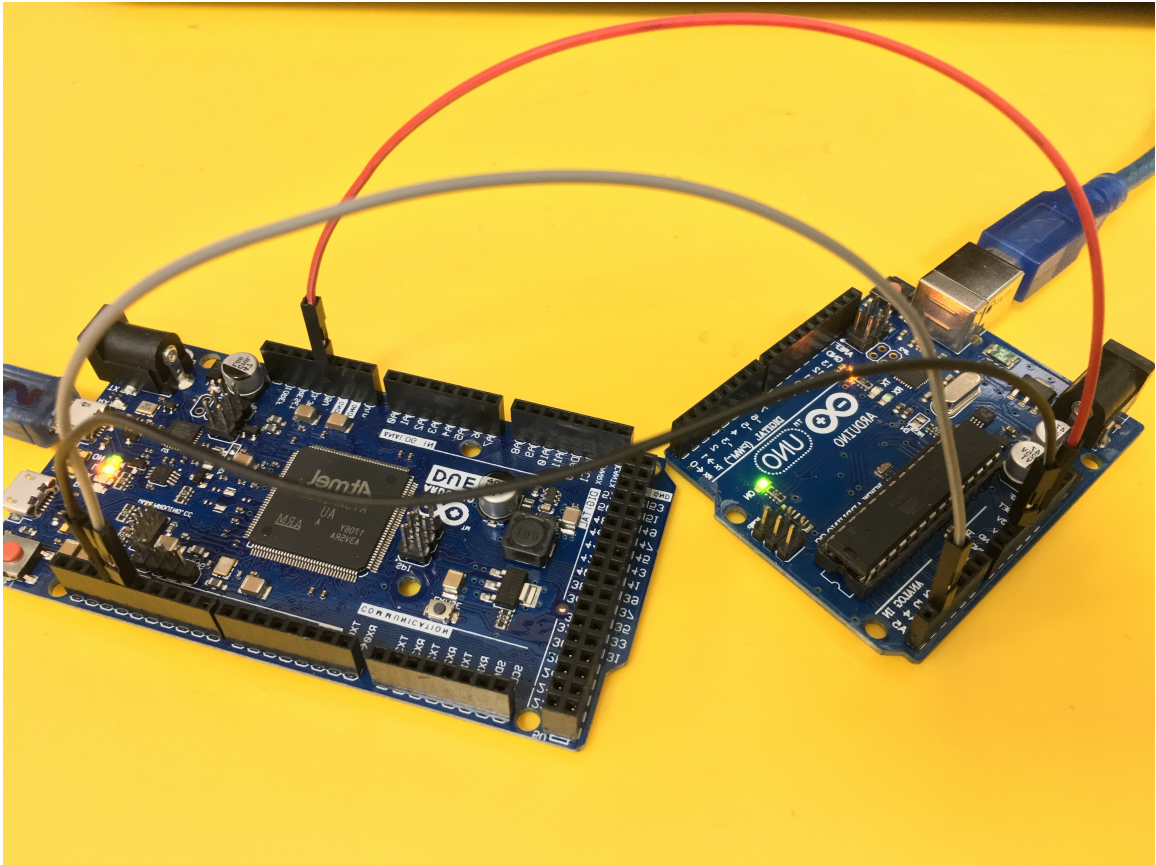


Figure A.6: One-O-One Arduino based PRBS generation and observation.

Case-1: we attached a Led with the output of Arduino to visualize the real-time blinking of Led which helps us understand system has no error during connectivity of Arduino with MATLAB/Simulink.

Case-2: In this case, we attached the output of Arduino with the oscilloscope (Rohde and Schwarz, Model no. RT A4004) and observed signal as shown in fig.A.5. (case-2). We have observed signal at different time scales, and R-T FFT of PRBS at any interval of time.

Case-3: The Arduino Due attached output (i.e., transmitting signal) with another Arduino (in this case is Arduino Uno) input at pin A0 to observe it on MATLAB/Simulink scope, the as shown in Fig. A.6.

Case-4: This is a lowest-cost scenario to generate and observe signal via one Arduino

Table A.1: Comparison of different N-Bit PRBS

| S/no. | PRBS Bits | Clock frequency | Testing time |
|-------|-----------|-----------------|--------------|
| 1     | 6-bit     | 10 Hz           | 127.80 sec   |
| 2     | 8-bit     | 40 Hz           | 31.88 sec    |
| 3     | 10-bit    | 160 Hz          | 7.97 sec     |
| 4     | 12-bit    | 640 Hz          | 1.99 sec     |
| 5     | 14-bit    | 2560 Hz         | 0.50sec      |

board, i.e., the output of Arduino is a route back to its input pins to observe signal formation. In this case, Arduino provides resourceful independence from the use of lab equipment, e.g., oscilloscope and function generator.

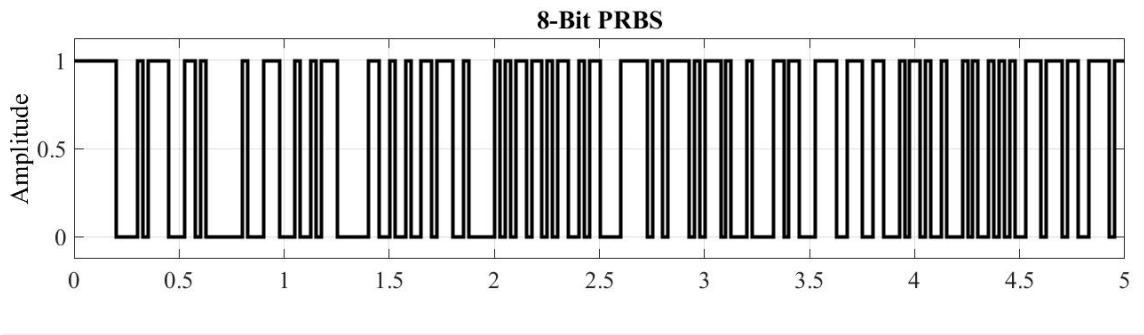


Figure A.7: Block functions used for Arduino input signal display on Simulink.

Here An Arduino board can be used as an oscilloscope and function generator mainly for the PRBS case. The PRBS has enormous applications in communication, control, power engineering but due to very high-cost function generator it was difficult to generate PRBS. Here we have solved this problem by providing designing of 8-bit PRBS and integrating with Arduino board. We have also observed Arduino as low-cost devices for multipurpose applications with high accuracy and less background knowledge of electronics.

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